

Universe24: the algebra of a reversible integer rule and the cube's group of 48

with its 24 rotations: exact theorems, the two slits and Bell's correlation under a declared credit, and the weak-field forms of a scalar pace

Alon Gonen

Independent researcher, Haifa, Israel

alon@defounder.ai ORCID 0009-0008-2701-4599

October 2026

Abstract

Abstract

We present a model of physics as one local rule in whole numbers on a cubic lattice, separating what it proves, fits and leaves open. Rule3 steps every Node from its present, past and six neighbours by one Euclidean division; we prove every run exactly reversible, a quadratic form conserved at fixed paces, and the rule covariant under the lattice's translations and the cube's group of 48. We formulate measurement as the meeting of the record run forward with the record read backward from a click; which meeting is realised is the one non-local element, and an observer sees only clicks. Families are bands of the rule, bodies its fixed points with a stability theorem. Gravity is a scalar pace, a level slowing a Node's clock once and its links twice; two features were fitted to light's bending and the perihelion; against the Newtonian potential the orbits measure, the post-Newtonian observables at the computed pair [2, 3] leave Einstein's by 25 to 33 percent and γ by 50, and the admissible regime, the closing gap, holds no computed body. The second-order form of Lorentz's factor follows unfitted; gravity's events, reading no well, run against GW170817. Without a run we derive the two-slit pattern and, under a credit rule declared to give it, Bell's correlation, $S = 2\sqrt{2}$, and the three-quantum GHZ correlation, Mermin's $M = -4$, where every local credit gives at most 2; the engine's gates read both exactly, blind first: the implementation gives what the algebra claims.

Keywords: integer lattice rule; reversible cellular automaton; measurement as a meeting of past and future; Bell's inequality; weak-field gravity; Lorentz invariance on a lattice

1 Introduction

The question. The question of this paper is whether one local, reversible rule in whole numbers on a lattice can give the clicks that nature shows, and with how little state, non-locality and randomness. It starts from one statement and never contradicts it (ASSUMPTION): the world is the meeting of the past with the future; a click is that meeting, the one measurement there is; and an observer sees only clicks (the words are defined at the end of this section). In that sense the model is the least computation we know of that gives the two slits and Bell's correlation under the declared credit; of nature's other clicks it gives the forms of Section 8 at the computed pair, with the findings of Section 10.3 against them. The model is a cubic lattice of Nodes on which one local rule in whole numbers acts at every Node in every interval of time. A Node holds a record's present and its past, two integer levels, and the rule makes the record's future from the two and from the six neighbouring Nodes. The rule is covariant under the lattice's translations and under the symmetry group of the cube, the 24 rotations and, with the reflections, the 48 signed permutations of the three axes; no boost is among them (Section 3.1). The lattice holds nothing but events that split: at every Node a level splits by the rule to the six neighbours. Run forward from an emitter, the split events are the future, every path at once; read backward from a realised click by the same rule, they are the past, and through two slits both pass both gaps. A click is where the two meet at a declared detector, and which meeting is realised is the one thing the model does not determine, the one non-local element of the description (Section 5.2). What appears as a wave is the future, what appears as a particle is the present, the click, and the track is the past (Section 5.1). Least is meant in three

named counts, the state per Node, the non-local elements and the random draws, among the models we know of; no theorem of minimality is claimed, and why nature is so we do not claim to know.

The rule in one line. Everything in the paper follows from one line of integer arithmetic, Rule3, together with one write and three readings. At every Node, in every interval and for every family of physical values [1, the line; the booking; the primitives]:

Rule3:	$w a_{\text{next}} + r' = \sum_{a \in \{x, y, z\}} R_a (\text{arr}_{+a} + \text{arr}_{-a}) + S a_{\text{now}} - w a_{\text{before}} + r, \quad 0 \leq r' < w,$ $w = 6 \text{ den } \Gamma^2, \quad R_a = 2 \text{ num } p_a^2,$ $S = 12 \text{ den } \Gamma^2 - 12 (\text{den} - \text{num}) p_0^2 - 4 \text{ num } (p_x^2 + p_y^2 + p_z^2);$	(1)
the one write:	$\text{a held part at a Node gains } \left(\sum_f w_f q_f(\text{Node}) + r \right) \text{ div } E, \text{ the remainder } r \text{ kept there;}$	
the readings:	$D_i = \text{now}_i^2 - \text{next}_i \text{ before}_i, \quad W_i = \text{re}_{\text{now}} \text{ im}_{\text{before}} - \text{im}_{\text{now}} \text{ re}_{\text{before}},$ $F_{ij} = \text{num} (\text{now}_i \text{ before}_j - \text{before}_i \text{ now}_j).$	

In the first line a_{now} , a_{before} and a_{next} are the family's level at the Node now, one interval before and one interval later: the present, the past and the future. $\text{arr}_{\pm a}$ is the arrival through the Port $\pm a$, the neighbour's level. r and r' are the remainder kept at the Node before and after the step, Γ the Node clock. $[\text{num}, \text{den}]$ is the family's pair, and $\cos \omega_0 = \text{num}/\text{den}$ is its rotation per interval at wave number zero. p_0 is the Node's clock pace and p_a its pace along the axis a ; both are the Node clock lowered by the field levels the family reads there (Section 3.4). div is the floor division. The second line is the only way a field's level is ever written: the sum over the sourcing families f of each family's own quantity q_f at the Node, weighted by w_f , plus the part's one remainder, divided by the part's wall E . The third line is what is read from a record and never written: its form D_i , which sources the fields; its Wronskian W_i , its charge; and its current F_{ij} into the Node i from its neighbour j , which a detector reads through its boundary in quanta of $W_c = 3 \text{ den } T$, with T the quantum's action.

What the paper contributes. We state the rule and prove what is exact about it, reversibility to the bit, the conserved form with its current, the momentum flux as the wave's own stress (Section 4); we formulate measurement as the meeting of the forward and the backward record at a declared detector, the one non-local element named (Section 5); we derive the families as bands of the rule and choose a universe with as few free numbers as the rule allows (Section 6); we show that bodies are fixed points, prove their stability under the dilation and state how they cluster (Section 7); we derive the weak-field forms of gravity and the second-order form of Lorentz's factor, each marked by what it rests on, in the model's own potential and in the one nature's tests read (Section 8); and we state the two slits and Bell as derivations of what each must give, neither experiment performed here (Section 9).

The path from the rule to the formulas. Every formula of the paper is reached from Eq. (1) by a chain of named steps, with every assumption named where it enters, so that each path is closed in the algebra; the supplementary material gives the algebra of each step (S.1 to S.25). Light's speed and its dispersion: the line at the vacuum's pace, which is the band (Section 3.3). Gravity, light's index, the bending and the delay: the line, the paces, with the one assumption that a field's level is read once into the clock and twice into the Link, the band at a pace, the index, the ray integral (Sections 3.4 and 8). The redshift: the band at wave number zero with the clock's pace. The fall, the Newtonian potential the orbits measure and the post-Newtonian parameters: the band at small wave number, the inertia, the rays (Section 8). Lorentz's factor: the exact band of a free record and the rotation at its moving centre, closed for every family, exact at the second order and the full factor as the family's gap closes (Section 8, (e)). From the Node to us: the line with its write, the fixed point and its functional, the stability theorem, the cloud and its fall, the clicks between bodies, the clusters and the observer as a cluster (Section 7). The count: the conserved form, the share, the invariant (Sections 4.3, 4.4 and 7.4). Light's birth, absorption and emission: the write line, the Wronskian, the resonance (Sections 5.6 and 7.4). Bell: the line's linearity, the parts' determinism, and the declared credit, the one assumption of that chain (Section 9.4). The two slits: the line stepped on the declared world (Section 9).

What is put in, and the three kinds of claim. The paper's claims are of three kinds, and the text keeps them apart. *Found*: what is proved or derived from the rule with no input chosen for it, the reversibility, the conserved form, the group's action, the bands, light's speed and dispersion, the fixed point of a body and its stability under the dilation, the redshift's band factor, the second-order form of Lorentz's factor, the universality of free fall for extended bodies, the two-slit pattern and the click formulas of Table 2 marked derived; the findings against nature are comparisons, listed in Section 10.3. *Fitted or declared*: the three choices below, the universe's rows, and the detector's declarations. *Held to be computable and not yet computed*: the closing-gap limit of the matter pair and the bodies at it, the moving fixed point of a body, the absorbed fraction with the body's write, and the two worlds' click counts beside nature's; and, outside the model under their own names, the hypotheses, a stepping Γ , a massless row that reads the content,

Table 1 The results of the paper, each with its kind, its fence and what it rests on. A result marked FITTED rests on an input chosen with that result known, so it shows consistency and is not a prediction; a calibration is a number taken from a measurement of nature; EXPERIMENT is a number the model does not fix, with the experiment named in the text. The fence says whether a formula is a reading of the lattice (GAMEBOARD) or a formula of what a detector reports (CLICKS).

Result	Kind	Fence	Rests on	Section
Every run is reversible to the bit	theorem	GameBoard	Rule3	4.1
The gate says MATCH on the two slits' and Bell's worlds	computed, a run	GameBoard	the implementation	4.1
The rule's acts form no product of levels	theorem	GameBoard	Rule3	4.2
A conserved form, its current, and the momentum flux as the wave's stress	theorem	GameBoard	Rule3	4.3
The count is the record's share; the total share is never negative	theorem	GameBoard	Rule3	4.4
Three exact rotations	theorem	GameBoard	Rule3	6.1
Light's speed $1/\sqrt{3}$, its dispersion, and no dispersion along the cube's diagonal	derived	GameBoard	Rule3 for the pair [1, 1]	3.3
A click is a meeting, credited by the shares; no collapse; no signal	assumption, derived	clicks	the meeting and the detector	5
A field's reach from its gap, Yukawa's form	derived	GameBoard	Rule3 for a gapped pair	6.4
The stability of a body under the dilation, to first order in the well	theorem	GameBoard	the body's functional	7.2
Free fall is universal for extended bodies of one family	derived, computed	GameBoard	the fixed point and the binding range	7.3
A body's clicks: no light from a steady rotation, light at a breathing body's beat, the resonance at $2\omega_b$, the count's drift	derived	clicks	Rule3 at the body's Nodes	7.4
A cluster: the three levels at a Node as the nesting, three kinds of binding by rank, the click as the only thing between clusters, the recursive hierarchy	assumption, derived	GameBoard, clicks	the universe is bound; Rule3	7.5
The gravitational redshift, $z = f(\omega_0) U$	derived	clicks	the clock's read	8
Light's bending $4U_b$, its delay and the perihelion's $6\pi U$, in the clock's potential	fitted	clicks	the Link read twice; the clock's second order	8
The same forms against the Newtonian potential the orbits measure: $\gamma_K = \text{den}/\text{num}$, $\beta_K = (\text{den} + \text{num})/(2\text{num})$, $1 + \alpha_K = \tan \omega_0/\omega_0$	derived, against nature at [2, 3]	clicks	the fall's formula	8
The second-order form of Lorentz's factor, at the family's own light speed, the full factor as the gap closes	derived	clicks	Rule3's band	8
The force between parallel moving charges	not derived under the rule	clicks	a hypothesis outside the rule	10.3
De Broglie's relation, the fall and Newton's pull	derived	GameBoard, clicks	the band and the field's rest	8
Newton's pull, the clock's G and Kepler's G	derived	GameBoard, clicks	Rule3 for [1, 1] and E_g	10.1
The two-slit fringe row and the lattice's arrival interval	computed	clicks	Rule3 on the declared world	9.2
Bell's $E = \cos 2(a - b)$ and $S = 2\sqrt{2}$, and $S(\rho)$ for unequal parts	fitted	clicks	the credit rule, the square of the meeting	9.4
Every local credit gives $S \leq 2$	theorem	clicks	Rule3 and a local credit	9.4
The vacuum content, 0	assumption	clicks	declared; GW170817 a finding	6.5
Tsirelson's bound $2\sqrt{1 + \rho^2}$ for the credit's E	derived	clicks	the credit rule	9.4
The binding range, 20 Links	calibration	GameBoard	the proton's 1,836 quanta	7.3
Gravity's divisor, the proton's count, the unit of charge, the matter pair's gap, the Links' wall	calibration, experiment	clicks	nature's measurements	6.6, 10.2
The findings against nature of Section 10.3: the counter's unit, the proton's size and nuclear binding, the neutron's sign, the waves' polarisations, light's anisotropy, frame dragging, the post-Newtonian observables at [2, 3], gravity's events against GW170817, the deceleration	derived, against nature	GameBoard	Rule3	10.3

and a binding read of the rotation's kind. Three choices were made with a known result in view, and we state them as inputs, not predictions. First, a field's level enters a Link's pace twice, a choice fixed by the bending of light; the bending and the delay of light are therefore consistency checks. Second, the clock's pace carries a second-order term, adopted with the perihelion's advance known; the perihelion is therefore a consistency check. Both hold in the clock's own potential; against the potential that Kepler's orbits measure, the same forms reach Einstein's only as the matter family's gap closes (Section 8). Third, a click's share is the square of the meeting, a credit rule chosen to reach nature's correlations; Bell's value $2\sqrt{2}$ follows from it and does not test it. Five numbers are nature's and not the rule's: the vacuum content, fixed by a measurement; the binding range and the proton's count, tied to each other, the count an input; gravity's divisor, bounded; and the unit of charge, a hypothesis (Section 10.2). The magnetic force between moving charges does not follow from the rule as it stands, (Section 10.3). Table 1 lists every result of the paper with its kind, its fence and its inputs.

How to read the paper. Every claim carries one of six marks. THEOREM: proved for every lattice. DERIVED: obtained from theorems by argument. COMPUTED: a number obtained from the rule's lines before any run. ASSUMPTION: a line held by decision and named where it is used. HYPOTHESIS: a line outside the model, under its own name.

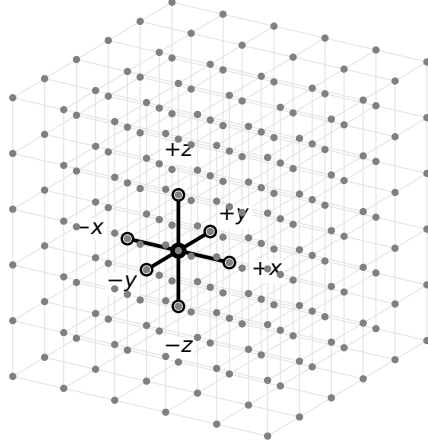


Fig. 1 The GameBoard in space: a cubic array of Nodes, here $6 \times 6 \times 6$, each joined to its six neighbours by Links through the Ports $\pm x$, $\pm y$, $\pm z$ (the faint lines); one Node and its six neighbours marked, the Nodes one interval away from it. A Node keeps each family's two levels and its remainders there, and nothing else. Drawn from the definitions; no run.

EXPERIMENT: a number the model does not fix, with the experiment named that would fix it; the paper performs none. A result that rests on an input chosen with that result known is also marked FITTED; it shows consistency, not a prediction. Every formula carries, beside its mark, its fence. GAMEBOARD: a reading of the lattice's own numbers that no instrument reports, a diagnostic. CLICKS: a formula of what a detector reports, the share Q_R/W_c on the forward orbit (Section 5.2) or a ratio of two such, the only kind that nature can be set beside. The algebraic steps of every derivation that the text states without its proof are given in the supplementary material (Online Resource 1), Derivations S.1 to S.25 in the order of the sections, cited as (S.*n*) where used. The runs of the implementation reported in this paper are the gates' (Section 3.7), the blind written first and the run identical to it, the engine's check of the algebra and no experiment against nature. The law's document, the implementation, the derivations with their checks and the text were worked out with a large language model used as a tool under the author's direction, and the author reviewed every statement; the Statements and Declarations at the end say how. The paper cites three documents of its archive by the names of their sections: the law [1], the engine [2] and the decisions [3].

The words. The physical location is a *Node*, and its complete local information is its *NodeState*. Nodes are joined by *Links* through six directional *Ports* per Node, and the lattice of Nodes is the *GameBoard* (Fig. 1). A local change is an *Event*, and there is one local rule, Rule3. A *family* is one kind of physical value with its pair and what it holds. Its *record* is its levels over the GameBoard, two per Node with a remainder. A family of quanta has a *dimension*, 1 for one real line and 2 for a plane of two lines; a part of a record, one level pair with its remainder, is what the implementation calls a line. A *body* is the set of Nodes on which a family's record stands bound; its quanta are the record's share there, and it has no law of its own. A *cluster* is bodies standing in one another's wells, bound by the same fields and clicking at one another; a stone is one. A *message* is a travelling record laid by the world, such as an emitter's packet of light. A *detector* is a region of Nodes declared in the file to report, and its *click* is one quantum credited to it from the current it saw through its boundary; the click is the one measurement. An *interval* is one step of the whole GameBoard. A family that holds a value for others to read is a *field family*, or *holder* [1, the objects][2, the words].

2 Related work

Reversible cellular automata and digital physics. The idea that the physical world is a discrete computation goes back to Zuse [4]; it was developed as reversible cellular automata by Fredkin, Toffoli and Margolus [5][6][7], as a programme by Wolfram [8], and as a deterministic foundation beneath quantum theory by 't Hooft [9][10]. Quantum cellular automata put a unitary step on a lattice, from the lattice Weyl, Dirac and Maxwell equations [11] and quantum lattice gases [12] to the structure theorems of the reviews [13][14] and the derivation of Weyl's and Dirac's equations from an automaton's symmetries [15]; the present rule's step is an integer recurrence on two levels, not a unitary on amplitudes, and quantum behaviour is sought in the clicks and not in the state. Rule3 is a second-order cellular automaton in the sense of Toffoli and Margolus, $a_{\text{next}} = f(a_{\text{now}}) - a_{\text{before}}$, reversible by construction [7]; what the model adds is an integer alphabet with one Euclidean division whose remainder stays at the Node, and the reading of

the automaton’s quadratic forms as physics. Feynman noted that a local classical computer reproduces no correlations beyond Bell’s bound [16]; the model agrees, and places the excess in the declared credit of the click rather than in the lattice (Section 9.4).

Time-symmetric accounts of measurement. Wheeler and Feynman described radiation as an interaction between the source and the absorber, with advanced and retarded fields together [17]; Cramer’s transactional interpretation has an offer wave forward and a confirmation wave backward, meeting at the absorber [18]; the two-state vector formalism evolves one state forward from the preparation and one backward from the outcome [19][20]; Price argued that a time-symmetric account is what Bell’s theorem leaves open [21], and Wharton and Argaman showed that every locally mediated reformulation of quantum theory must let some mediating parameters depend on measurement settings in their future [22]. The model belongs to this class: the backward reading from a realised click depends on the detector’s setting, and nothing else does. It differs in that both directions are the same integer line on a lattice that is exact and reversible to the bit, that the product of the two directions is a reading and not a wave, and that the choice among the meetings is named as the one non-local element rather than derived. The record is an ontic state in the terms of [23].

Bell’s theorem and local models. Bell’s inequality bounds every local account of a pair’s correlations [24], in the form that experiments test [25], and a local model, a joint distribution and the inequalities are one condition [26]. The model’s local credits give $S = \sqrt{2}$ and $S = 2$, as they must (Section 9.4); its credit by the contrast reaches the quantum correlation only by discarding pairs, Pearle’s road [27], below the efficiency bounds of Eberhard and of Garg and Mermin [28][29], which the loophole-free tests of 2015 close for nature [30][31][32]. Superdeterminism [10][33] the model does not take: its settings are declared.

Discrete gravity and its tests. Regge calculus [34] and causal sets [35] discretise the geometry of space-time itself; here the lattice is never curved, and the weak-field forms are read from the coefficients (Section 8). In the parametrised post-Newtonian framework [36] the two fitted choices give $\gamma = \beta = 1$ with $\alpha_U = 0.063$ in the model’s own potential and $\gamma_K = \text{den}/\text{num}$, $\beta_K = (\text{den} + \text{num})/(2\text{num})$ against the potential the tests read. The universality of free fall, which the model derives for extended bodies of one family, holds in nature to 10^{-13} on the ground [36] and to 10^{-15} in orbit [37]. A lattice breaks Lorentz invariance at order one unless its terms are tuned [38], and the bounds are reviewed in [39]; the paper’s answer is the closing gap of Section 8.

3 The model

This section states the model: the lattice, the families, the rule, the paces through which one family acts on another, the one write, the readings and the order of acts in an interval [1, the GameBoard and the families; Rule3; the primitives].

3.1 The GameBoard

The GameBoard is a box of $N_x \times N_y \times N_z$ Nodes. Two Nodes that differ by one step along one axis are joined by a Link, and each Node has six Links, one through each of its six Ports, $\pm x$, $\pm y$ and $\pm z$. Each axis is declared periodic, open or folded. Beyond an open face no Node lies and a level there is read as 0; a folded axis has one layer, and its two Ports return the Node itself. An inner face is a plane of Nodes declared closed except for declared gaps, which is how a wall with slits is made. A face may also be declared receding (ASSUMPTION): the box then grows by a layer of zeros whenever a nonzero level reaches the Node next to the face, so no event ever meets the face, and a run on such a box is a run on the unbounded lattice. On a receding face the fields build their rest behind the front by Rule3 itself, as Huygens’ construction on the lattice, and nothing reflects; the box grows and nothing stretches, so a receding face gives no cosmological redshift [1, the objects; the unbounded board][3, the receding face].

The group and its action. The symmetry group of the cube has 48 elements, the 24 rotations and, with the reflections, the signed permutations of the three axes, O_h ; with the translations it is the space group of the simple cubic lattice. It acts on the state space by permuting the Nodes and their Ports and carrying every level pair with its Node; the time reversal exchanges a record’s present and its past and carries the step to its inverse (Theorem 1); and on a plane record, whose two parts obey one line, a rotation of the two parts by an angle with a rational cosine and sine, $(p, q)/r$ with $p^2 + q^2 = r^2$, carries the rational solutions of the line without its division to solutions and leaves the Wronskian unchanged; the integer step’s three shears (Section 5.5) are a reversible map of the integer state and not this rotation. Rule3 commutes with the space group’s action, since its line reads the six Ports symmetrically and names no axis: the law is covariant, and a world turned by an element of the group runs as the turned world. Under the group the form D is a scalar, even under the time reversal; the current F a vector, odd under it; the Wronskian W a pseudoscalar; and the click a scalar, since it names a region and an interval and never a Node or an axis. No boost, no rotation beyond the cube’s 24 and no scaling is in the group, so Lorentz’s group and the continuum’s rotations are

no symmetries of the lattice, and what the paper finds of Lorentz's factor it finds as a property of the clicks (Section 8) (THEOREM for the covariance and the readings' kinds; GAMEBOARD; S.4) [1, the direction; the four acts][3, the one algebra].

The NodeState holds the model's own numbers and nothing else: for every family and every part, two levels and one remainder, and for every part of a field family, the remainder of its one write. A count, a current, a charge or a tension is a reading computed from these numbers when it is needed; no register, table, formula or random draw is kept at a Node [1, the objects][3, nothing at a Node but the law's numbers].

3.2 Families and records

A family is declared by two keys: its pair [num, den] and its shape. A family of quanta, such as matter or light, declares its dimension, 1 for one real line or 2 for a plane of two lines, and holds nothing. A field family declares its sources, the form, the form with the three tensions, or the Wronskian, which fix its parts, one real line per source, with its divisor E_s and, for the massless field, its rest; a field sourced by the form or the tensions holds the content, which slows the clocks of the families that read it, and a field sourced by the Wronskian holds the sign, the charge's field. A family's record is its levels over the GameBoard, at every Node a level now and a level one interval before, with a remainder; a family of dimension 2 lays two level pairs (re, im), two solutions of the same line, and only a plane carries a charge (Section 5.5). A body is a connected set of Nodes on which a family's record stands bound, and a message is a travelling record laid by the world [1, the objects].

3.3 Rule3

At every Node, in every interval, every family steps by the first line of Eq. (1). The six arrivals through the Ports, each weighted by its axis's coefficient R_a , the Node's own level times S , minus w times the level before, plus the remainder the Node kept, are summed as one whole number and divided once by the wall w ; the quotient is the next level, and the remainder, $0 \leq r' < w$, stays at the Node and never travels. Nothing is rounded away, and the step back is the same division, which is why every run is exact and reversible to the bit (Theorem 1). The name comes from the three axes: the line has one coefficient per direction serving that direction's two Ports, the six Ports and the wall's factor 6 are the three directions twice, and for light the next level is the sum of the six arrivals over 3, minus the level before [1, the line].

$$a_{\text{next}} + a_{\text{before}} = \frac{\text{num}}{3 \text{ den}} \sum_a (\text{arr}_{+a} + \text{arr}_{-a}), \quad (2)$$

whose rotation at the wave vector $\mathbf{k}$ is $\cos \omega = \frac{\text{num}}{3 \text{ den}} \sum_a \cos k_a$. At wave number zero $\cos \omega_0 = \text{num}/\text{den}$, and for the massless pair [1, 1] along one axis $\cos \omega = (2 + \cos k)/3$, so $\omega = |k|/\sqrt{3}$ as $k \rightarrow 0$ (DERIVED, GAMEBOARD; S.1). Light's speed at long wavelength is therefore $1/\sqrt{3}$ Links per interval, below the causal bound of one Link per interval, and its group speed falls at shorter wavelengths as $1 - 0.083 k^2$ [1, the lattice constants]. In nature's names the vacuum's line is the discrete Klein-Gordon equation exactly: with $S_6 = 6 + \Delta_{\text{lat}}$, $a_{\text{next}} - 2a_{\text{now}} + a_{\text{before}} = c_K^2 \Delta_{\text{lat}} a_{\text{now}} - m^2 a_{\text{now}}$ up to the remainder term $(r - r')/w$, with $c_K^2 = \text{num}/(3 \text{ den})$ and $m^2 = 2(1 - \text{num}/\text{den})$, the rest rotation as the mass (c_K is the equation's coefficient; the band's speed c_m of Section 8, (e), is $\omega_0/(3 \tan \omega_0)$, the two one as the gap closes); its slow limit at small wave number is Schrödinger's equation with the inertia $m^* = 3 \tan \omega_0$ and, in a content U , the potential $\omega_0 f(\omega_0) U$ of Section 7.4 (DERIVED, GAMEBOARD); and $m^* c_m^2 = \omega_0$ exactly for every pair, with $c_m^2 = \omega_0/(3 \tan \omega_0)$ of Section 8, (e): the model's $E = mc^2$ at the family's own light speed, read in three click numbers, nature's its small-gap limit, the departure $1 - \omega_0^2/3$ (DERIVED, CLICKS; S.18, S.21). At the vacuum's pace Rule3 is the leapfrog scheme of the Klein-Gordon equation, whose dispersion is the known one [40]; what the law claims of it is four things: one rational [num, den] fixing mass and speed together, the equation exact and not an approximation, its departures predictions with bounds (Section 10.2), the integers with the remainder, and the paces (DERIVED).

3.4 The paces: how a field acts on a record

A field acts on a family by changing the family's paces at a Node, the coefficients of Rule3 there:

$$p_0^2 = (\Gamma - c)^2 + c^2, \quad p_a = \Gamma - 2c - t_a, \quad c = \sum_{\text{reads}} \text{weight} \times \text{the read family's time level}, \quad (3)$$

Here c is the content the family reads at the Node, the sum over its reads of the weight times the read field's level, and t_a is the axis content from the fields' tensions on the axis a . A positive weight makes a hollow, an attraction; a negative weight makes a hill. A family with no reads steps at $p_0 = p_a = \Gamma$. The pace is the clock's slowing: a record at

a Node of pace p rotates and moves as a record at Γ does, with its intervals p/Γ as long. Every family of quanta reads every holder of the content at the weight 1, so $x = (L_b + L_g)/\Gamma$ with L_b the binding field's level and L_g gravity's; a family of dimension 2 also reads the holder of the sign plainly, and light, the holder's own record, reads the holders of the content as every family does and not the holder of the sign; and a family sources what it reads at the weight it reads with, so every pair of families that read each other is reciprocal [1, the paces; the reads are the held row's pair's].

Two choices fixed with a known result. The level enters the clock's pace once and each Link's pace twice. Nothing in Rule3 forces the factor two on the Link: it was chosen so that light passing a body bends by twice the clock's share, as nature's light does [36] (ASSUMPTION, FITTED). The clock's pace carries the second-order term of a slowing composed unit by unit, $p_0 = \Gamma(1 - 1/\Gamma)^c$, whose square to second order is $(\Gamma - c)^2 + c^2 = \Gamma^2(1 - 2\Phi + 2\Phi^2)$ with $\Phi = c/\Gamma$; this term was adopted with the perihelion's advance known, the clock to first order giving 7π times the potential per orbit and to second order Einstein's 6π (ASSUMPTION, FITTED). They are inputs of the model, and Section 8 says where each is used [1, the clock once and the Link twice; the clock's second order is the composition][3, the Node clock and the ruler].

The band at a pace. From the coefficients, the rotation of a record at a Node of clock pace p_0 and Link paces p_a is (DERIVED, GAMEBOARD; S.1)

$$\cos \omega = 1 - \left(1 - \frac{\text{num}}{\text{den}}\right) \left(\frac{p_0}{\Gamma}\right)^2 - \frac{\text{num}}{3 \text{den}} \sum_a \left(\frac{p_a}{\Gamma}\right)^2 (1 - \cos k_a), \quad (4)$$

The mass term scales with the clock's square and the band term with the Link's. Two readings follow. First, the clocks shift alike: at $k = 0$ every massive band's rest rotation obeys $1 - \cos \omega_b = (p_0/\Gamma)^2(1 - \cos \omega_0)$, one factor for every family, so two bound bodies at one Node shift their clocks alike. Second, along a Link every wave's speed falls with $p_a = \Gamma - 2c - t_a$, so light is delayed and bent by twice the clock's share. The line is not conformal: a light wave at a fixed wave number scales with $(p_a/\Gamma)^2$ and a bound mode at rest with $(p_0/\Gamma)^2$. At a Node of content $x = L/\Gamma$ light's band is $[2 - 4(1 - 2x)^2, 2]$ in $2 \cos \omega$, and at long wavelength $\omega = (1 - 2x) |k|/\sqrt{3}$: light's speed is $(1 - 2x)$ times its vacuum speed, an index $n = 1/(1 - 2x)$ with $n - 1 = 2x$ (S.2). A record whose rotation lies outside the band at a Node is evanescent there and reflected whole; light meets a total mirror where $p_a < \Gamma \sin(k/2)$ [1, the band at a pace; the rows against nature, (b)].

The guard. A pace must be positive and below an edge P , with $P^2 = 2 \text{den} \Gamma^2 \text{div} (\text{den} + \text{num})$, which is Γ^2 for light: beyond P the mode at wave number π on every axis grows without bound, and at or below P it stays bounded (THEOREM, GAMEBOARD). The guard is the stability condition of Courant, Friedrichs and Lewy for the explicit second-order scheme that Rule3 is at the vacuum's pace [40]; it is checked once, on squares, over the whole initial state, and never during a run. At $R_a = 0$, a content of $\Gamma/2$, a Link closes: that is a horizon. The clock never stops there, since $(p_0/\Gamma)^2 = (1 - \Phi)^2 + \Phi^2 \geq 1/2$ for every content: at the horizon the Links close while the clock's pace is $1/\sqrt{2}$ of the vacuum's, a massive band's rotation there $\cos \omega = 1 - g/2$, 0.70 of ω_0 at $[2, 3]$, and a Node whose three Link paces are 0 rotates at $2 \cos \omega = 1 + \cos \omega_0$ (DERIVED, GAMEBOARD; S.3). Such a frozen Node is cut from its six neighbours: no current crosses its Links, its share is not a number, and it stands in no instrument's front, so the horizon is dark. The guard's bound, $|2 \cos \omega| \leq 2$ at the band's edges, excludes exponential growth and admits the edge itself, where a linearly growing solution exists, and it is stated for $\text{num} \geq 0$, a negative numerator's lowest rotation lying at $k = 0$ (S.3). At a negative content light's zone-corner mode grows, so light refuses a content below 0, and the vacuum's content is 0 (Section 6.5) [1, the guard; the frozen Node has no share].

3.5 The write, the readings and the interval

The one write. Every write into a field's level is one act, the second line of Eq. (1): at a Node in each interval a held part gains $(\sum_f w_f q_f + r) \text{div } E$, one division per part with one remainder r carried at that Node. Here q_f is the sourcing family's own quantity at the Node: its form D_f for a holder of the content, its Wronskian W_f for the holder of the sign, or its tension $T_{aa,f}$ for an axis part. w_f is the weight with which that family reads the written field, and E is the part's wall, $E_s T$ for the time part and $E_s W_c$ for an axis part. The sources are summed in the numerator and never divided apart, so no remainder stands per sourcing family [1, the primitives][3, the one algebra].

The readings. A reading, which we also call a booking, is a bilinear form of the levels with declared coefficients; it writes nothing on the GameBoard. The law reads three, the third line of Eq. (1). The form $D_i = \text{now}_i^2 - \text{next}_i \text{before}_i$, the present against the product of the future and the past, is $B^2 \sin^2 \omega_b$ at every interval for a standing record of peak B ; its quotient over T , the well, sources the fields. The Wronskian W_i is the charge (Section 5.5). The current $F_{ij} = \text{num}(\text{now}_i \text{before}_j - \text{before}_i \text{now}_j)$ into the Node i from its neighbour j is antisymmetric and is the Link term of the conserved form (Theorem 3); a wave travelling from i to j has $F_{ij} < 0$. A static record has $\text{now} = \text{next} = \text{before}$,

so its form and its current vanish: a static field carries no quanta and no detector sees it, by the arithmetic and not by a clause [1, the booking; the conserved form].

The interval. Every read is of the values at the interval's start; Rule3's line reads one Link per interval, and nothing in zero time; the interval as the implementation composes it, the step, then the tensions read from the stepped record, then the hold's write, has a dependency radius of three Links, finite and bounded, and larger than light's $1/\sqrt{3}$ (Section 4.3). The interval has four acts in order: (i) the read, each family's paces from the field levels; (ii) Rule3 on every record; (iii) the readings, including each record's six currents at the levels the step started from, its tension, form and Wronskian, and each detector's report, the net current into its region through the region's front boundary; (iv) the write of each field from its summed sources. Light is born at (iv), when a rotating body's varying Wronskian is written into the holder of the sign (Section 5.6) [1, the interval][2, the interval].

3.6 The postulates

The model rests on seven postulates [1, the postulates]: (1) the world is Nodes and Events, an Event a level stepped by Rule3 with its remainder, a click a detector's report, and nothing else happens; (2) a Node holds bounded local information, read from itself and its six neighbours, and nothing of the past is kept; (3) consistency is local and causal, one Link per interval for Rule3's line, the composed interval reaching three Links (Section 4.3), with one exception, the credit of a click by the shares over every declared detector, the one non-local element, not derived; (4) the causal speed is one Link per interval; (5) every physical calculation is on bounded integers, with no floating-point number, no root and no random draw in the law; (6) a measurement is a detector's click, the net current into a declared region through its front boundary over its window, never a Node; (7) the GameBoard, where no one measures, is deterministic and reversible to the bit, and the detectors' clicks are the only thing claimed to represent nature.

3.7 The implementation

The implementation, Universe24 [41][2], is the law's implementation and nothing else: a GameBoard of Nodes, every family's NodeState at every Node, and one interval of acts in the law's order, each a call of Rule3 on whole-board arrays of integers at a declared width, every neighbour read through a Port. It holds no number of physics, no formula, no family's name and no flag, every physical value coming from the run's files; a run is headless, writes one click line per family, region and interval, the measurement, beside field lines that are diagnostics, and runs for results only after the gate of Section 4.1 says MATCH [2, the engine; the NodeState; the output; the gates]. The gates' runs, the blind written first and the run identical to it: the two slits, $N = 278$ against the law's real line's 277.8 over the whole passage (the Huygens blind 273); Bell, $S = 478/169$ with the marginals $1/2$; GHZ, the eight shares, every pairwise E 0 and $M = -4$; and the back-in-time gate MATCH on every world (COMPUTED, runs of the implementation, the engine's check of the algebra and no experiment; CLICKS for the counts) [1, the rows against nature, (g) and (h); the GHZ gate]. To reproduce: `build_world.py` in a world's folder lays it, `tools/run_inputs.py` runs it and `tools/back_in_time.py` checks the inverse, as the engine document's section on running a world states [2, how to run a world]. The paper reports the gates' runs and derives the rest, for two reasons. The first is the method: the implementation is built as the law's lines and nothing else and is held to them by its tests and by the gate, so a run of a declared world computes what the algebra of that world states and adds no information where the algebra is closed; every result of the paper is reached in the algebra and marked so, and a run is the engine's check of the same line, needed where the algebra is not closed, the two worlds' click counts and a body's clicks beside nature's, which stand in the paper as what each must give. The second is the means: the project runs on one computer, which sets the order in which the declared worlds are run, not what they must give. The implementation and the documents are public in the repository named in the Statements and Declarations.

4 Exact properties

This section proves what holds exactly on every lattice: the reversibility of the rule, the kind of function its acts can form, the conserved form with its current and its momentum flux, and the count as a reading of the record.

4.1 Reversibility

Let σ be the direction, +1 forward and -1 backward; let div and mod be the floor division and its remainder in $[0, w)$; and let ρ be the remainder carried in. With

$$u = \sigma \left(\sum_a R_a \text{arr}_a + Sx - wy \right) + \rho, \quad z = \sigma(u \text{ div } w), \quad \rho' = u \text{ mod } w, \quad (5)$$

the forward call with $(x, y, \rho) = (a_{\text{now}}, a_{\text{before}}, r)$ gives $(z, \rho') = (a_{\text{next}}, r')$, and the backward call with $(x, y, \rho) = (a_{\text{now}}, a_{\text{next}}, r')$ gives $(z, \rho') = (a_{\text{before}}, r)$.

Theorem 1 (the direction). *The backward call returns (a_{before}, r) bit for bit; the reversibility of every step needs no second function.*

Proof. Forward, $\sigma = 1$, the line is the definition of `div` and `mod`. Backward, $\sigma = -1$: write Y for the right side's first two terms, $\sum_a R_a \text{arr}_a + S a_{\text{now}}$. Then $u = w a_{\text{next}} - Y + r'$ and $z = -\lfloor u/w \rfloor = \lceil (Y - w a_{\text{next}} - r')/w \rceil$. From the forward line, $Y - w a_{\text{next}} - r' = w a_{\text{before}} - r$ with $0 \leq r < w$, whose ceiling divided by w is a_{before} exactly; and $\rho' = u - w \lfloor u/w \rfloor = w a_{\text{before}} - (Y - w a_{\text{next}} - r') = r$. Checked bit for bit on 20,000 random cases against a forward and a backward function written separately [1, the direction]. $\square$

The whole run therefore goes back: N intervals forward and N back return every array of every family bit for bit, provided every number an act read at an interval's start is one the state after the interval still holds (THEOREM, by construction from Theorem 1). The implementation checks this as a gate, running a world forward and back and comparing every array at every interval, MATCH or the first interval and Node that differ; on the two worlds of Section 9 it reports MATCH, on the two slits' world over 40 intervals and, with its far face receding, over 130, and on each of Bell's four worlds over 100 intervals, every record of every family bit for bit, the writes' remainders among them (COMPUTED, a run of the implementation; GAMEBOARD, a diagnostic and no measurement) [1, the inverse; the objects][2, how to run a world][3, the Node knows no family]. The condition rules out one kind of read: a coefficient read from a Node's own state across a step cannot be recovered at the step back. This is why a record's rotation sense is never read at a Node, and why the holder of the sign is read according to the record's shape, a plane or a line, and not according to a sense (Section 5.5) [1, the interval][3, the board goes back in time, as a gate; there is no sense].

4.2 What the acts can form

Rule3's line is one act, and declared coefficients make it four: (a) the read, the line with its six arrivals; (b) the one-Node step, the line with the arrivals declared 0, which rotates a level by Chebyshev's recurrence; (c) the division, the line with $R = S = 0$, Euclid's division with the remainder kept; and (d) the load, a declared integer in the numerator, which is how a count enters a level [1, the four acts].

Theorem 2 (what the acts can form). *Every composition of the four acts is a piecewise-linear function of the levels with integer slopes: a sum of linear forms and of floors of linear forms, composed. No such function is quadratic in the levels.*

Proof. Each act is a linear form of the levels or the floor of one; a floor of a linear form is piecewise linear with integer slopes; sums and compositions of such functions are of the same kind. The second difference of such a function in any one level is 0 almost everywhere, while the second difference of a product $\text{now}_i \text{before}_j$ in the pair $(\text{now}_i, \text{before}_j)$ is 1. So nothing bilinear comes from the acts alone. $\square$

So every product of two levels is a reading and never a write, and the three readings of Section 3.5 are the only products the law takes. A line enters the law only if it passes three tests: it is generic (one primitive with declared integers, no family's name), vector (an act on the whole state, with no root and no floating-point number) and local (its own record and the six neighbours, nothing kept at a Node beyond the law's numbers). A line that needs more is a hypothesis under its own name. No act of the interval takes a square root; the integer square root appears only when a world is laid, as the fixed point of the division act $x \leftarrow (x + n \text{div } x) \text{div } 2$ [1, the three tests of every rule][3, the six verbs; the root leaves everywhere].

4.3 The conserved form and the momentum flux

The read matrix $2 \text{num } p_i^2$ is not symmetric where the paces differ, so the conserved form carries the weight $1/p_i^2$ on its Node terms and the plain current on its Links:

$$E = \sum_i \frac{w(\text{now}_i^2 + \text{before}_i^2) - S_i \text{now}_i \text{before}_i}{p_i^2} - 2 \text{num} \sum_i \text{now}_i S_6(\text{before})_i. \quad (6)$$

Theorem 3 (the conserved form). *At constant paces for which a positive Node weight d_i with $d_i R_a(i) = d_j R_a(j)$ on every Link exists, as it does when the three Link paces at each Node are equal ($d_i = 1/p_i^2$) and when each axis's pace is uniform in space ($d_i = 1$), E with the weights d_i is exact under the line without the rounding; with paces both anisotropic and varying in space no such weight exists in general, and the exact statement is the share identity Eq. (10)*

with its anisotropy term, E then drifting; with the rounding each Node's remainder adds at most one unit of the level per interval to the walk of E . Its Link term, the plain current $\text{num}(\text{now}_i \text{ before}_j - \text{before}_i \text{ now}_j)$ through the Link ij , is the current of Rule3's own conservation law.

Proof. Without the rounding the line is $\mathbf{a}_{\text{next}} + \mathbf{a}_{\text{before}} = \mathbf{M} \mathbf{a}_{\text{now}}$ with $\mathbf{M}$ the matrix whose diagonal is S_i/w and whose entries on the six Links are R/w . Let $\mathbf{D}$ be the diagonal matrix of the weights $1/p_i^2$; then $\mathbf{DM}$ is symmetric iff $d_i M_{ij} = d_j M_{ji}$ on every Link, $d_i R_a(i) = d_j R_a(j)$: with the three Link paces equal at each Node the weight $1/p_i^2$ cancels $R_a = 2 \text{num } p_i^2$ on both sides of every Link, and with each axis's pace uniform in space $\mathbf{M}$ is symmetric as it stands; with paces both anisotropic and varying in space the products of the ratios $R_a(i)/R_a(j)$ around a square of Links must be 1 for a weight to exist, which fails in general. For a symmetric $\mathbf{K} = \mathbf{DM}$ the quantity $Q(\mathbf{x}, \mathbf{y}) = \mathbf{x} \cdot \mathbf{D} \mathbf{x} + \mathbf{y} \cdot \mathbf{D} \mathbf{y} - \mathbf{x} \cdot \mathbf{K} \mathbf{y}$ satisfies $Q(\mathbf{a}_{\text{next}}, \mathbf{a}_{\text{now}}) = Q(\mathbf{a}_{\text{now}}, \mathbf{a}_{\text{before}})$: with $\mathbf{a}_{\text{next}} = \mathbf{M} \mathbf{a}_{\text{now}} - \mathbf{a}_{\text{before}}$, $\mathbf{a}_{\text{next}} \cdot \mathbf{D} \mathbf{a}_{\text{next}} - \mathbf{a}_{\text{next}} \cdot \mathbf{K} \mathbf{a}_{\text{now}} = \mathbf{a}_{\text{before}} \cdot \mathbf{D} \mathbf{a}_{\text{before}} - \mathbf{a}_{\text{before}} \cdot \mathbf{D} \mathbf{M} \mathbf{a}_{\text{now}}$, which is the other side. Multiplied by w this is E up to the Link term's sign convention [1, the conserved form]. $\square$

E is a diagnostic. Where the paces move, the total form $D = |\mathbf{a}_{\text{now}}|^2 - \langle \mathbf{a}_{\text{next}}, \mathbf{a}_{\text{before}} \rangle$ changes exactly by $D(t+1) - D(t) = \langle \mathbf{a}_{\text{next}}, (\mathbf{M}(t) - \mathbf{M}(t+1)) \mathbf{a}_{\text{now}} \rangle$, which is 0 at fixed uniform paces and negative where a well deepens under a standing record (THEOREM, by substitution; GAMEBOARD) [1, the conserved form].

The momentum flux is the wave's own stress. The momentum density along the axis a at the Node i is the current per axis, $P_a(i) = (F_{i,i+a} - F_{i,i-a})/\text{num}$, and the line without its division gives exactly, at constant paces, the identity below; the integer step adds the remainder term $\delta_i \Delta_a(\text{now})_i - \text{now}_i \Delta_a(\delta)_i$ with $\delta = r - r'$ and Δ_a the centred difference, which the implementation's test includes (THEOREM for the line, GAMEBOARD),

$$w [P_a(t+1) - P_a(t)] = \sum_b R_b [G_{ab}(i-b) - G_{ab}(i)], \quad (7)$$

$$G_{ab}(i) = \text{now}_i (\text{now}_{i+a+b} - \text{now}_{i-a+b}) - \text{now}_{i+b} (\text{now}_{i+a} - \text{now}_{i-a}),$$

with $G_{ab}(i)$ the stress reading on the b -Link $(i, i+b)$, the levels now alone. The proof substitutes $w \text{ now}' = \sum_b R_b (\text{now}_{i+b} + \text{now}_{i-b}) + S \text{ now}_i - w \text{ before}_i$ into $w P_a(t+1)$: the S term cancels, the w term is $w P_a(t)$, and the R term is the difference of the fluxes. The tension on the axis a at a Node, the part a field reads into the Link's pace and the source of a field's axis part, is the momentum flux averaged over the Node's two a -Links,

$$T_{aa}(i) = -\text{num} (G_{aa}(i) + G_{aa}(i-a)) \text{ div } 2, \quad (8)$$

$$G_{aa}(i) = \text{now}_i (\text{now}_{i+2a} - \text{now}_i) - \text{now}_{i+a} (\text{now}_{i+a} - \text{now}_{i-a}),$$

and its reach is two Links along the axis, since the difference at the far end of the neighbour's a -Link enters it: it is a reading of the Links' differences and not of one Node's six neighbours, the one booking of the law that reaches beyond them; read from the stepped record, as the implementation composes the interval, the axis part written at a Node depends on the state three Links away at the interval's start, so the interval's dependency radius is three Links and not one, finite and bounded, and whether the write of a field's axis part meets the local test as stated is open (Section 10.3). A plane wave of amplitude b at the wave number k has $G_{ab} = -2b^2 \sin k_a \sin k_b$, so the flux along the wave is positive, as a wave's pressure is (S.7); a packet of count c and velocity $\mathbf{v}$ has $c v_a v_b$, the stress of dust. With the sign as written a wave's stress deepens the content around it, so light gravitates by its pressure too. The off-diagonal parts and the vector part, which Rule3 never reads, are not written [1, the tension; the final unification][3, the tensor part is the record's own stress].

4.4 The count is the record's share

A family's count of quanta is a reading of its record and no line of its own. At every Node the record's share is

$$e_i = 3 [w (\text{now}_i^2 + \text{before}_i^2) - S_i \text{ now}_i \text{ before}_i] \text{ div } \left(2 \sum_a p_a^2 \right) - \text{num now}_i S_6(\text{before})_i, \quad (9)$$

the conserved form's Node term over the mean of the three Link paces squared, less the plain Link term, read in quanta over W_c as the nearest whole number. Read with the vacuum's coefficients instead of the Node's own paces, the same expression is the vacuum's share, the unit of the tables of Section 7. Nothing is kept at a Node beside the record and the write's remainder: no count, no remainder of a count and no sense. Over one step of Rule3 the share's rational value, the weighted Node term before its floor, changes by exactly the six currents into the Node, plus a term that vanishes where the three Link paces are equal, plus Rule3's own rounding (THEOREM for the rational identity,

GAMEBOARD); the integer share of Eq. (9) adds the reader's own floor, a residual strictly between -1 and 1 per Node and level pair, distinct from Rule3's remainder:

$$e_i(\text{next}, \text{now}) - e_i(\text{now}, \text{before}) = \sum_j F_{ij} + \text{num}(\text{next}_i - \text{before}_i) \sum_a \frac{(3p_a^2 - P^2) \text{arr}_a(\text{now})_i}{P^2} - \frac{3(\text{next}_i - \text{before}_i)(r'_i - r_i)}{2P^2}, \quad (10)$$

with $P^2 = \sum_a p_a^2$, arr_a the two arrivals of the axis a summed, and r, r' Rule3's remainder before and after the step. The identity is derived in S.5 and checked by a test of the implementation up to one reader's floor per Node and level pair. So over any region whose three Link paces are equal, the total share changes only by the currents through the region's boundary, the remainder terms and the readers' floors (THEOREM, GAMEBOARD) [1, the count is the record's share; the share's change is the currents]. A formulation that moves whole quanta between Nodes by a division of the current, and the sign by a division of the Wronskian's current, would put two more numbers at every Node that no act of Rule3 reads back; the law does not carry it (ASSUMPTION) [3, the count's line and the sense's line leave the law]. Where the paces move, as under a breathing body, the share read at the new paces drifts with no current crossing any Link; the drift's formula is Eq. (13) of Section 7.4, a diagnostic [1, the paces' own change moves the share with no current].

The signed integers. The levels are signed integers, the currents are signed by their direction and the Wronskian by its sense, and a click is one whole quantum, never negative. The total share of a closed or periodic GameBoard is non-negative for $|\text{num}| \leq \text{den}$ (THEOREM, GAMEBOARD): with $\mathbf{n}$ and $\mathbf{b}$ the vectors of now and before and S_6 the six-neighbour sum, a symmetric operator whose spectrum lies in $[-6, 6]$, $\sum_i e_i = 3 \text{den} (|\mathbf{n}|^2 + |\mathbf{b}|^2) - \text{num} \mathbf{n} \cdot S_6 \mathbf{b} \geq 3(\text{den} - |\text{num}|)(|\mathbf{n}|^2 + |\mathbf{b}|^2) \geq 0$, with equality for a static uniform level of light and, on an even periodic board, for its checkerboard, now $= (-1)^{x+y+z}$ with before $= -\text{now}$, whose shares are 0 (S.6). One Node's share can be negative at a lay's edge, and a region's by at most half a level-square per boundary Port. Over the whole passage of one forward packet the inflow accumulated at a region is non-negative up to the rounding, since the lay's current is forward at every Link, so the credit by the shares is well defined; where a window is cut inside a passage the instrument credits $\max(s_R, 0)$ (DERIVED, CLICKS) [3, the law's integers are signed].

5 Measurement: the click as a meeting

The click is the meeting of the future with the past at a declared detector, and no act of the GameBoard. This section states what is on the lattice, the meeting, the detector, what a click leaves behind, the charge and light, and what locality costs [1, the click; the reading].

5.1 Events, and no waves

On the GameBoard there are events and amplitudes. An event is a level split by Rule3 to the six neighbours; an amplitude is the level Rule3 produces at a Node, the sum of the split events there; and the record is the amplitudes over the GameBoard. In our world there are clicks, the detectors' reports, and nothing else. In the two slits the emitter's events pass both gaps, every path of the future at once; the fringes are where split events from the two gaps meet at the screen with the same sign. The past is made of the same split events: read backward from the realised click by the same line, Rule3 at -1 , they form the past's tree, which also passes both gaps back to the emitter. What stays on the GameBoard, the amplitudes no click realised, is not in our world (ASSUMPTION) [1, no waves on the event board][3, no waves on the event board].

5.2 The click is the meeting of the future with the past

Assumption 1 (the click is a meeting, the law and not a hypothesis). A record runs on the GameBoard in both directions by the one line: forward from the emitter, the possibilities, the future; backward from the realised click, what happened, the past, Rule3 at -1 . A click is where the two carry the same signal at a declared detector; its share is the product of the two directions at the region, the form $D = \text{now}^2 - \text{next} \times \text{before}$, the now against the future times the past; which meeting is realised is the draw, the one non-local place. The detector only asks whether the future and the past met at it.

Computationally, by the linearity of the line, the backward reading from a realised click equals a sum over the forward records, so the meeting adds a reading of what a click is and no mechanism of choice; which meeting is realised stays the draw by the shares, the one non-local place.

In plain words (Fig. 2): a click happens when the past meets the future; the click is the now; the detector only asks whether the future and the past met at it; and a click is created only at their meeting. This is a line of the model and not a hypothesis [3, the click is the meeting of the future with the past].

The click, algebraically. Let X be the state space: at every Node, for every part of every family, one integer level pair and the write remainders, and nothing else. The step S_t is Rule3 over the interval with the paces read from the fields at its start; it is a bijection of X , exact by the division with its remainder; the line without its division is linear in the levels at fixed paces, the linearity the backward reading uses, while the integer map is not affine, the floor of $4/3$ of 1 and of 2 not adding to that of 3; its forward orbit is the future and its inverse orbit the past. The readings D , F and W are bilinear forms on X , covariant under the symmetry group of Section 3.1 and never written. An instrument R is a declared region with its front Ports and its coefficients, which define Q_R : for one line a quadratic form of the levels, the inflow through its front Ports accumulated over the window; for a pair, the window's sum of the squared cross-side sum over the parts, quadratic in the joint amplitude, the product of the two sides' levels, and so quartic in the levels. A click is the pair (R, t) , the credit of one whole quantum to the instrument R at the interval t , drawn with the probability $s_R / \sum_{R'} s_{R'}$, where $s_R = Q_R(x_t) / W_c$ is the instrument's form on the forward orbit over the quantum's measure of a current. In the meeting's words, the inverse orbit from the credited state coincides with the forward orbit on the realised path, and the share is the bilinear form's value at the coincidence. The click writes nothing to X : it is the one map in the model that is not a bijection, from the vector of shares to a unit vector, and it is a scalar under the group, so no click names a Node. Every observable is Q_R / W_c on the forward orbit or a ratio of two such. The window is the whole passage of a packet, or the credit floors at 0 (Section 4.4); a Node whose three Link paces are 0 stands in no instrument's front and is never clicked (ASSUMPTION for the definition, DERIVED for the precisions; CLICKS) [3, the click's algebraic definition].

One form for every click. At the credit the parts of a record are read with the detector's coefficients, each part paired with the same part through the common origin, summed and squared; the share is that density, and the draw is by the shares. The two slits are the case with nothing to pair: light is one real line, and the share is the record's own inflow. Bell's gate is the case through the common origin: a pair born as one event is two trees with one root, and the backward reading from the first click passes the root and continues forward on the partner's tree (Section 9.3). In the implementation the joint share is computed from the forward records alone, by the linearity of the rule; no backward Rule3 is stepped, and the gate of Section 4.1 shows that the backward reading exists to the bit (DERIVED, CLICKS).

No duality. What is called the particle as a wave is the future: the record run forward by Rule3 with every possibility, which has not yet met the past. What is called the particle as a particle is the present: the click, one whole quantum credited at one region in one interval. The particle's track is the past: the chain of realised clicks, and the same record read backward through them. There are not two things but one record read in two directions, and the point where the two readings coincide is what appears as a particle. After a click no wave vanishes and nothing collapses: the record goes on, the click is a credit, and the past has grown by one click (ASSUMPTION) [3, the wave is the future, the click is the present, the track is the past].

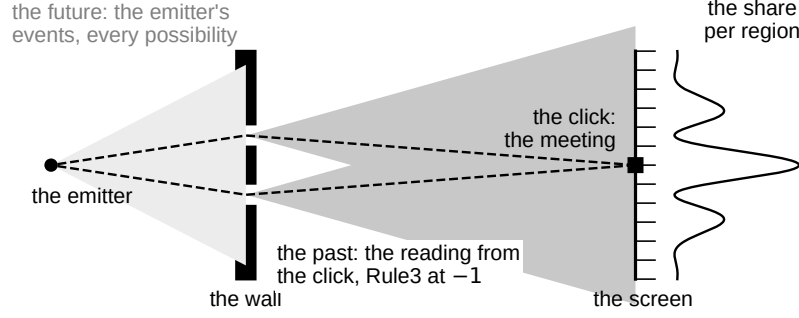
5.3 The detector is a declared instrument

Assumption 2 (the detector's declaration). A detector is one connected region of Nodes declared in the file, never one Node, at least half the wavelength of the family it reports across the beam (q/p Links for the wave $[p, q]$, 4 at $\lambda = 8$); a narrower region, a single Node or a region in pieces is refused by name at load, and the depth along the beam is not gated. It is opaque: backed by a receding face or by the wall's own face, so that what enters it leaves the declared board and never returns, with no act at its Nodes and no log. Its click is its report, per interval and family: the net current into the region through its front boundary Ports, the Ports leading in from the declared board, summed in integers, with the region's name and the family and never a Node; one wall $W_c = 3 \text{ den } T$ of the inflow over a passage is one quantum, so a screen saw N quanta where its regions' inflows sum to $N W_c$. The credit: one quantum to one detector, drawn by the shares of the regions' inflows, read outside the GameBoard over every declared detector of the world, one credit per quantum and never two. The credit is the one non-local place of the description and no line of the law.

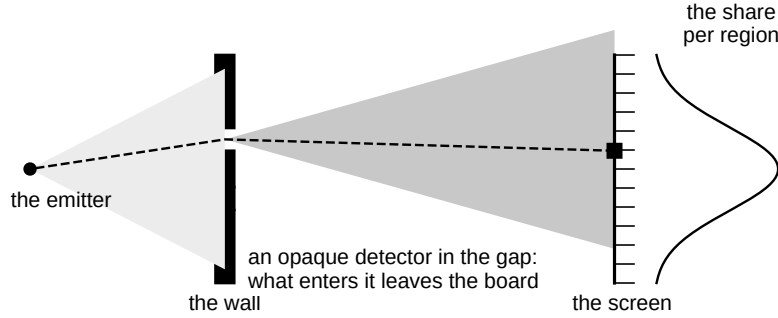
Why a region: a record confined to one Node contains every wave number at once, $\Delta x \Delta k \geq 1/2$ on the lattice as in nature, and the finest structure a family's amplitudes carry is half its wavelength; so no click names a position finer than a region, and every per-Node number is a diagnostic [1, the click is the detector's report; no click names a Node] [3, no click names a Node].

What the model determines and what the instrument declares are separate. What produces a click is deterministic: the meeting, the current of the amplitudes through a region's boundary. What is not known is which of the possibilities becomes the click: when one possibility is spread over several detectors, the GameBoard gives the shares, no line of the

(a) two gaps: the share alternates



(b) one gap closed: one gap's envelope



(c) a pair: the meeting through the root

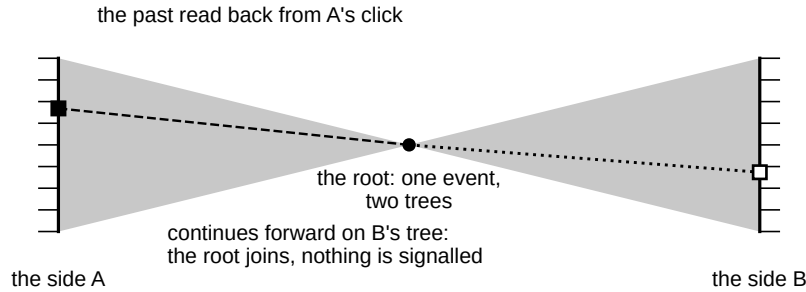


Fig. 2 The click as the meeting of the future with the past, schematically (ASSUMPTION; drawn from the definitions, no run). (a) The two slits: the emitter's events spread forward through both gaps, every possibility at once (the grey fans, the future); from a realised click at a region of the screen the same line is read backward, Rule3 at -1 , through both gaps to the emitter (the dashed lines, the past); the share along the screen, the product of the two directions, alternates. (b) One gap closed by an opaque detector: its events leave the declared board; the backward reading meets one gap and the share is one gap's envelope. (c) A pair, two trees with one root: the backward reading from A's click passes the root and continues forward on B's tree (the dotted line); the root joins and nothing is signalled.

law gives which detector fires, and the instrument draws. Exclusivity, one click per quantum, and Born's rule, the credit's chance equal to the intensity's share [42], are therefore declarations of the instrument and not findings of the model, as nature's detector efficiency and dead time are [1, what produces a click is not known]. The uncertainty relation is not a separate rule: it is the detector's three declarations, a region of at least half a wavelength, a credit per interval and one draw per quantum (ASSUMPTION) [3, the uncertainty relation is not a rule of its own]. What each declaration costs: a region of n Nodes reads a fringe of period Λ with the factor $\sin(\pi n/\Lambda)/(n \sin(\pi/\Lambda))$, 0.90 for four rows on a fringe of sixteen; a frequency is a click rate over a window of N intervals, read to $\Delta\omega \geq 2\pi/N$; and N quanta drawn by the shares give each region $Np_i \pm \sqrt{Np_i(1-p_i)}$, so a pattern is in the histogram and never in one click (DERIVED, CLICKS; S.8). The credit by the shares is done by the reader of the output and never by the GameBoard [2,

the output].

5.4 The click ends nothing

Assumption 3 (the click ends nothing). The click is a detector's reading, local to its boundary; the record goes on after it and nothing ends anywhere. What a detector absorbed leaves the declared board through the face that backs it; the other possibilities go on splitting and leave through the receding faces, events that were not realised, in no click. The ending of a record at a click by the detector is decided against and is no hypothesis of the law.

This is the model's claim and not a deficiency: perhaps nature is local. No signal passes, since nothing at one detector changes the amplitudes at another. A measurement needs no collapse: the record never ends, the click is a reading, and the backward reading from the click writes nothing [3, the click ends nothing]. A bound body can act as an instrument in the same form: the future is light's forward record at its boundary, the past is the body raised by one quantum read backward to the source, and the meeting is the same frequency, the resonance of the body's rotation with the light's; emission is the same meeting reversed, and a body at rest and symmetric clicks nowhere (DERIVED, CLICKS; Section 7.4). The body's write back waits on the fixed point with the write [3, a Node of the board is read only through a detector][1, the rows against nature, (y)].

5.5 The sign is the rotation sense

A family of quanta laid as a plane carries its record as two level pairs, (re, im), stepped by one call of Rule3 with one set of coefficients, each with its own remainder; a field's parts stay real. The charge density at a Node is the reading

$$W_i = \text{re}_{\text{now}} \text{im}_{\text{before}} - \text{im}_{\text{now}} \text{re}_{\text{before}}, \quad (11)$$

the Wronskian of the two pairs, which the line without its division conserves exactly with the conserved form's weights, under Theorem 3's condition on the paces (THEOREM, since **DM** is then symmetric; GAMEBOARD; S.9). The integer step adds to each part the remainder term $\epsilon = (r - r')/w$, $|\epsilon| < 1$, so the integer W changes at a Node by $\epsilon_{\text{re}} \text{im}_{\text{now}} - \epsilon_{\text{im}} \text{re}_{\text{now}}$ per interval, below $|\text{re}_{\text{now}}| + |\text{im}_{\text{now}}|$ against a W of the order of the levels' product, and not zero in general: at the vacuum's pace of [2, 3] the parts (1, 3) with zeros before step to (1, 4) and W from 0 to -1 (COMPUTED, GAMEBOARD; S.9); an exactly conserved integer charge is open (Section 10.3). Its local change is the Link current num $(\text{im}_i \text{re}_j - \text{re}_i \text{im}_j)$, and its total over the GameBoard is invariant under any change of the paces, since the bracket $\text{im}_{\text{now}}^\top \mathbf{M} \text{re}_{\text{now}} - \text{re}_{\text{now}}^\top \mathbf{M} \text{im}_{\text{now}}$ vanishes for every symmetric $\mathbf{M}$ (THEOREM, GAMEBOARD). A real record ($\text{im} = 0$) has $W = 0$, so light, the real record of the holder of the sign, is neutral; a rotating record has W of either sign by its sense [1, the sign is the rotation sense; the sign's count is the record's own].

Who reads the holder of the sign is decided by the family's dimension and by no name (ASSUMPTION, GAMEBOARD): a family of dimension 2 has a Wronskian and reads it, a family of dimension 1 has $W = 0$ and does not, and a neutral body and a charged body of the same pair are two rows of the file with the dimensions 1 and 2. A field does not read its own level, so light does not read the holder of the sign and a static charge does not bend light. A reader's own sense never enters its coefficients, since a coefficient read from a Node's own state would break the exact inverse (Section 4.1); so the holder of the sign acts on its readers by the act its row declares: the plain read of its level into the paces, **pace**, under which like and unlike senses are not distinguished, or **rotation**, the holder's level over Γ as the angle θ turning the two-part record in its own plane, $e^{-i\theta} z_{\text{next}} + e^{i\theta} z_{\text{before}} = (S z_{\text{now}} + \sum_a R_a (z_{+a} + z_{-a}))/w$, three exact integer shears inverted bit for bit with θ read at now alone, under which a record of one sense turns faster by θ and the other slower, so that like and unlike senses are told apart by the record's own sense; which sense moves apart is one bit the model does not fix, declared by nature's click that like charges repel, with the engine's convention written beside the line, $e^{i\theta} z_{\text{next}} + e^{-i\theta} z_{\text{before}}$ on the same right side, the opposite sign of θ , which the declared bit absorbs, the shears' exact angle $2 \arctan(L/2\Gamma)$, L/Γ at small angles, and the magnetic force follows as the Link's phase, Peierls' coupling, with its blind not run (ASSUMPTION for the declared bit, GAMEBOARD; Section 10.3) [3, there is no sense; every family has a dimension; the rotation round, built][1, the sign holder rotates the two-part record; the rows against nature, (f)].

5.6 Light is born by the write

Light is the holder of the sign's own record: one record whose time part is a real level pair, stepped by Rule3 at the paces of its reads, so that light is bent by the content, and written by the one write from the Wronskian of the records that read it. A static rotating body writes a static level, Coulomb's field, and radiates nothing. A breathing or moving body writes a varying source at its Nodes, the field carries it away at one Link per interval, and those spreading events are the light (DERIVED, CLICKS). Its count is the share of that record, read like every family's. A detector reads light's current through its boundary in quanta of W_c , and the travelling events of the other fields, which in gravity's case

nature calls a gravitational wave, enter a detector the same way. A body's quanta that leave it stay its family, and no quantum changes family at a crossing. A body reads the holder of the sign at its own Nodes, so a breathing body reads the light it writes and its pace answers, the radiation reaction, in the same loop [1, light is born by the write][3, light is born by the write].

5.7 The price of locality

Three consequences of the model's locality, stated against nature's clicks [1, what is open, item 4; the rows against nature, (g) and (h)]. (i) *Exclusivity and Born's rule are declared*: a quantum spread over several detectors is a set of shares, the model gives the shares and the instrument gives the click. (ii) *The which-way measurement*: an opaque detector in one gap absorbs that gap's events, whether or not it is credited, and the screen sees the other gap alone, a closed slit by construction; a which-way measurement that marks the path without absorbing it is beyond the instrument the model has. (iii) *Bell*: a pair is one laid record, a common cause, and every local credit of its phase gives $|S| \leq 2$ (THEOREM, CLICKS; Section 9.4) [24][25]; the meeting through the pair's root gives $E = \cos 2(a - b)$ and $S = 2\sqrt{2}$ with every quantum counted, while nothing is signalled, and this rests on the credit rule, a declaration of the model (ASSUMPTION, FITTED; CLICKS) [3, the click is the meeting of the future with the past].

- (i) *Exclusivity and Born's rule are declared*. A quantum spread over several detectors is a set of shares of the current at each; the model gives the shares and the instrument gives the click. A dilute packet, under one quantum per Node, builds its pattern click by click by the declaration.
- (ii) *The which-way measurement*. An opaque detector in one gap absorbs that gap's events, whether or not it is credited, and the screen sees the other gap alone, with no fringes: a closed slit by construction, in nature too. A which-way measurement that marks the path without absorbing it is beyond the instrument the model has.
- (iii) *Bell*. A pair is one laid record, a common cause, and its phase is a shared classical variable inside Bell's hidden variable [24]. Every local credit of that phase gives $|S| \leq 2$ (THEOREM, CLICKS; Section 9.4) [25]. The meeting through the pair's root gives $E = \cos 2(a - b)$ and quantum theory's $S = 2\sqrt{2}$ with every quantum counted, while each side alone reads one half at every port and nothing is signalled; this rests on the credit rule, the square of the meeting, which is a declaration of the model and not a derivation (ASSUMPTION, FITTED; CLICKS). The loophole-free test of 2015 that measured the sum found $S = 2.42 \pm 0.20$ [30], the value quantum theory itself predicts for that experiment's state fidelity, $2\sqrt{2}$ being the ideal state's [3, the click is the meeting of the future with the past].

5.8 The formulas of clicks

Every observable of the model is Q_R/W_c on the forward orbit or a ratio of two such (Section 5.2). Table 2 collects the formulas of clicks the model gives, each with its status and the section that derives it; the bodies' formulas rest on the body as the fixed point of its binding field (Section 7), and no row rests on a run [3, the click's algebraic definition; the mathematics of the bound bodies and how it bears on the clicks].

6 The families and the universe

A family is a band of the one rule and not a separate law. Its row in the universe file holds only what the rule and the geometry do not fix: its pair [num, den], with $\cos \omega_0 = \text{num}/\text{den}$, and its shape: for a family of quanta its dimension, for a field its sources with its divisor. A pair written over the GameBoard's one denominator Γ makes a mass one integer m , with $\cos \omega_0 = m/\Gamma$ and the wall $6\Gamma^3$; a reduction to lowest terms is a reading and never an act of the loader, so such families share one resolution of their remainders [1, the families from the rule; the mass is one integer; the wall is one].

How a family enters. A family enters the world as one row of the universe file with two keys and nothing else: its pair, and its shape, which for a family of quanta is its dimension, 1 or 2, and for a field its sources, the form, the form with the tensions, or the Wronskian, with the field's divisor E_s and, for the massless field, its rest. Its rotation at rest, its band, its coefficients and its wall, $w = 6 \text{den} \Gamma^2$, follow from Eqs. (1) and (4); its parts are its sources' count, one real line per source; its reach is set by its gap (Section 6.4); its reads follow from its dimension (Section 3.4), and it sources what it reads at the weight it reads with. Every record clicks, a field's included, since no row declares whether it clicks. A new family is one more row of the file and no code. Table 3 gives the rows of the universe, and Section 6.6 the numbers of the world [1, a family's declaration; the families from the rule][2, how to add a family].

Table 2 The formulas of clicks, each with its status: derived from the rule's line, computed from it, a declaration of the model, a hypothesis, or a number for experiment to fix. Every row is a formula of what a detector reports, CLICKS, and no row rests on a run. U is the clock's share and U_K the Newtonian potential the orbits measure (Section 8).

The formula	Status	What the model gives	Section
Born's rule, the count at a region	derived	$N_R = Q_R/W_c$, the share of the inflow through the region's front over the window; the pattern is in the histogram, $Np_i \pm \sqrt{Np_i(1-p_i)}$	5.2, 5.3
The dead time	declaration	not derivable: the shares credit a region any whole number of quanta in one interval, and W_c over the inflow's rate is the mean spacing of the clicks and not a least one	5.3
The two-slit row	derived; computed	the near-field minima at the rows 15 and 33; the screen's inflow peaks at the interval 68.0 against a continuum wave's 61.8	9.2
The arrival and its spread	derived; computed	$t = L/v_g$ with $v_g = (\text{num}/(3 \text{ den})) \sin k / \sin \omega$; the click's duration $\sigma_t = \sqrt{\sigma_x^2 + (\omega'' L \sigma_k / v_g)^2} / v_g$ from the band's curvature ω'' , 4.1 intervals at $k = \pi/4$ over 34 Links	9.2
Bell's pairs	the credit declared (Born's rule for a pair); E and S derived under it	$E = \cos 2a \cos 2b + \rho \sin 2a \sin 2b$ and $S = \sqrt{2}(1 + \rho)$, 478/169 at the integer settings for equal parts	9.4
GHZ, three quanta	the same declared credit; E_3 and M derived under it, read by the engine's gate exactly	the same credit on four lines: $E_3 = \cos 2(a + b + c)$, every pairwise E 0, Mermin's $M = -4$ where every local credit gives $ M \leq 2$	9.4
The clock's click rate	derived	$1 - f(\omega_0)U$ for a bound rotation, $f = 2(1 - \cos \omega_0)/(\omega_0 \sin \omega_0)$, and $1 - U$ its small-gap limit; a light clock $1 - 2U$; the second-order form of Lorentz's factor at the body's own light speed	7.4, 8
The bending's clicks	fitted in U ; derived in U_K	$4U_b$; $(1 + \text{den}/\text{num}) 2GM_K/(c^2 b)$ against the Newtonian potential the orbits measure	8
The absorption line	derived (the read's side); computed when the fixed point with the write is	parametric at $2\omega_b$, the growth $\epsilon/(4 \sin \omega_b)$ and the width $\epsilon/\sin \omega_b$; the pair line at $\omega_i + \omega_j$ of which it is the degenerate case; the transfer line at $\omega_j - \omega_i$, its rate waiting on a second check	7.4
Emission	derived	a steady rotation and a uniform motion write no light; a breathing body writes at its beat, at the first order in the breathing's depth	7.4
The which-way sum	derived	the continuity of the one inflow through the two fronts: a region behind one gap reads that gap's share and nothing of the fringe	5.7
The uncertainty relation	declaration, with its inequalities	$\Delta y k \geq \pi$ from the region, $\Delta \omega \geq 2\pi/N$ from the interval, $\sqrt{Np(1-p)}$ from the draw	5.3
No signalling	derived	$P(A+) = (\cos^2 a + r^2 \sin^2 a)/(1 + r^2)$, independent of b for parts of any amplitudes	9.4
A body's count in clicks	derived	the share summed over the body's Nodes, what a full absorber credits; under a moving well it drifts by Eq. (13)	7.4
Magnetism	not derived under the plain read; derived under the holder's rotation, not run	$(1 + 2\beta^2) F_C$ under the plain read, the wrong sign; F_C/γ under the rotation at the wall $\text{den } T$, $\lambda = 1/c^2$: 0.995 at 0.1c and 0.866 at 0.5c	10.3

6.1 The exact bands, and the third band

The line of one Node at rest, $a_{\text{next}} = 2 \cos \omega_0 a_{\text{now}} - a_{\text{before}}$, is exact in integers only where $2 \cos \omega_0$ is an integer. So the rule carries exactly three exact rotations, $2 \cos \omega_0 = 1, 0$ and -1 , with the periods 6, 4 and 3 and the pairs $[1, 2]$, $[0, 1]$ and $[-1, 2]$ (THEOREM, GAMEBOARD). A record on an exact band at rest in the vacuum leaves no remainder in its rest rotation, so it never clicks and only turns. The exact bands are exact only in the vacuum: at a Node of pace p_0 the band's rest rotation is $2 \cos \omega = 2 - 2g(p_0/\Gamma)^2$, an integer at $p_0 = \Gamma$ alone, so inside a well a record on an exact band leaves a remainder and clicks like every family's [1, the exact bands; the exact bands are exact only in the vacuum].

The bands $[0, 1]$ and $[-1, 2]$ are allowed by the rule and unused by the universe [1, the third and the colours, closed as absent].

6.2 The rule's own universe

The universe of the model is the one whose every number is the rule's own: the pairs from the integers 1, 2 and 3 of the three exact rotations and the three axes, every divisor 1 except gravity's, and the remaining free numbers named. Its

Table 3 The rule’s own universe, its five families as the file `examples/events/rule.json` holds them (DERIVED from the two keys of a row, its pair and its shape): each family with its pair [num,den] (in brackets, written over the Node clock $\Gamma = 6,000$), what it holds at its divisor, its shape (for a field its sources, one real line each; for a family of quanta its dimension), what it reads, and its free number. A sixth family, the pair family of light, two real lines never summed, is declared by Bell’s world and not by this file (Section 9).

Family	Pair	Holds, at the divisor	Shape	Reads	Its free number
gravity, the time part and the three tensions	[1, 1] ([6000, 6000])	the content, E_g (at least 10^6 at $\Gamma = 6,000$, a click bound from the two slits’ screen; 1,000 in the bodies’ worlds) the sign, 1	1 + 3 real lines, sourced by the form and the three tensions	nothing	E_g , the hierarchy; its rest c_{vac} , 0 by nature’s clicks
the charge, the holder of the sign, whose quanta are light, born by the write	[1, 1]		1 real line, sourced by the Wronskian of its readers	the holders of the content, at the weight 1; never its own level	none
the binding field, the strong force; its own events have the gap	[2400, 2401]	the content, 1	1 real line, sourced by the form	nothing	the pair, its range $R = 20$ Links, tied to Γ by the proton
$\cos \omega = \text{num}_b / \text{den}_b$ the bound charge (the polarisation), the exact band	[1, 2] ([3000, 6000])	the content, 1	1 real line, sourced by the form	nothing	none; a field of under one Link’s reach
matter, the scalar; a quantum that leaves a body is a message of matter	[m , Γ] ([4000, 6000])	nothing	dimension 1, one real line; a charged body of the same pair is a second row of dimension 2, a plane	the holders of the content, $x = (L_b + L_g) / \Gamma$; the plane also the holder of the sign	m , the mass

rows stand in Table 3, as the file `examples/events/rule.json` holds them (DERIVED from the two keys of each row) [1, the families from the rule; the rule’s own universe][2, the input].

The universe’s integers are the Node clock $\Gamma = 6,000$, a multiple of 6 so that every pair is exact over it; the quantum’s action $T = 2^{15}$; and the integers’ width, 63 bits, a bound and not physics. Γ and T are units, as c and $\hbar$ are in nature, and no line of Rule3 fixes T , since the rule is linear and the scale of its form is free. Matter reads gravity and the binding field plainly, $x = (L_b + L_g) / \Gamma$, and reads the bound charge; charged matter, a second row of the matter pair at the dimension 2, reads the holder of the sign as well; light reads the same holders of the content and is bent by them, and does not read the holder of the sign, its own field. The bound charge’s field is read by neither of the implementation’s two worlds and leaves the universe file unless a use is found (ASSUMPTION) [3, nothing that is not Rule3 remains]. Every field of the file is a record stepped by Rule3 at its pair and written by the one write; no field is laid by hand during a run, and a world’s starting rest is the fixed point of the division act, computed once before the first interval (ASSUMPTION) [1, the final unification][3, two units and the bounds; nothing that is not Rule3 remains].

6.3 What each family carries, and how gravity enters as a clock

What a family carries is set by what Rule3 reads and what a record can write, not by a name (DERIVED, GAMEBOARD). Rule3 reads two things of a Node and no third: the clock’s pace p_0 , one number, and the Link’s paces (p_x, p_y, p_z), the three directions of the rule’s vector. So a field that acts on the paces carries at most one time part and three axis parts, 1 + 3. A record of one real line, neutral matter or light, has a form D and a stress and no Wronskian; a record of a plane, charged matter, has a Wronskian W as well; a field is always real, one line per source, since a plane’s own W would be its own source. The shapes follow with no choice: *the scalar*, one real line, for matter, light, the binding field and the bound charge; *the plane* for charged matter; *one plus three*, the time part with the three tensions, for gravity, sourced by the form and by the diagonal of the record’s own stress (Section 4.3); and *two real lines never summed at a Node* for the pair family of light in Bell’s world. The law as it stands takes no vector part of a field and no off-diagonal stress, since Rule3 reads them nowhere; for a real record no current’s read exists within the three tests, and gravity’s readers are real records, so the law has no frame dragging of neutral bodies, a finding against nature (Section 10.3) [1, the final unification; the tension][3, every family has a dimension].

How gravity enters: a slowed clock, and no curved space. There is no metric, no force and no law of motion in the model. Gravity enters in three steps. (i) *The write*: at every Node in each interval every family of quanta writes its form over the action, $D_i \text{ div } T$, into gravity's field over the divisor E_g , by the second line of Eq. (1). (ii) *The field*: gravity's field, $[1, 1]$, steps by Rule3 at the pace 1, so around a static source it rests at Poisson's solution, $3G(r)s$, which is $3s/(4\pi r)$ far away; a change travels at one Link per interval, retarded, and the exact inverse is kept. (iii) *The read*: every family reads the field's level c at its Node into its own coefficients, Eq. (3), the clock's pace once and each Link's pace twice. That is the whole of gravity. A slower clock is what a deeper well is: every massive band's rest rotation slows by the same factor, so two bound bodies at one level shift their clocks alike and a far detector reads the emitter's slow clock. What nature calls curved space is here the Link's pace: along every Link through the well a record's speed falls by the level twice, so light's index is $n = 1/(1 - 2x)$; no space is bent. The factor two on the Link and the clock's second order are the two fitted choices of Section 3.4 [1, the paces; the clock once and the Link twice][3, the Node clock and the ruler].

6.4 The reach of a field, and the binding field

A field's rest solves $\text{num} \Delta^2 a - 6(\text{den} - \text{num})a = -3\text{den} \sigma$, its line with the source once an interval, with Δ^2 the six-neighbour second difference. Outside a body its rest therefore falls by $e^{-\kappa}$ per Link with $\cosh \kappa = 3\text{den}/\text{num} - 2$, the family's own band at zero rotation: the gap is both the mass of the field's quanta and the inverse reach of the field, one number (DERIVED, GAMEBOARD). $[1, 1]$ has no gap, and its rest is Poisson's over the whole GameBoard. $[1, 2]$ has $\cosh \kappa = 4$ and falls by 0.127 per Link, a contact well of $1/\kappa = 0.48$ Links' reach. A pair with num just under den has $\kappa^2 = 6\text{den}/\text{num} - 6$ to second order, a range $R = 1/\kappa = \sqrt{\text{num}/(6(\text{den} - \text{num}))}$ Links, which is Yukawa's form [43] (S.11) [1, the well].

Assumption 4 (the binding field and gravity's divisor, decided with a reservation). The well of a record, $D_i \text{ div } T$, is laid into two fields. The binding field, a family with a gapped pair $[\text{num}_b, \text{den}_b]$, $\text{num}_b < \text{den}_b$, at the divisor 1, stepped by Rule3 at its pair as the massless field is and started at its rest, which with the gap is a sum with a range, $R = \sqrt{\text{num}_b/(6(\text{den}_b - \text{num}_b))}$ Links and $e^{-r/R}/r$ beyond it: it binds the bodies, it is the model's strong force; its own travelling events have the gap $\cos \omega = \text{num}_b/\text{den}_b$, a quantum with a mass whose Compton length is R , Yukawa's own line [43] and no input. The massless field $[1, 1]$ at the divisor E_g declared by the file: gravity, long-ranged and weak by E_g . A family reads both levels into its pace at the weight 1.

The first pair is [2400, 2401], with $R = 20$ Links, and the field's quantum has the rest frequency 0.029 per interval. The pair is tied to Γ by the proton: a cloud of 1,836 quanta is bound under $R = 20$ and not under $R = 10$ or 15 (Section 7.3). In worlds with bodies E_g is declared large enough that gravity does not enter a body's binding, 1,000 in the computations of Section 7, a world's own number under the bound of Section 10.2. Nature's E_g is about 3×10^{42} , and the hierarchy of gravity and electricity is the file's one number, as G is nature's. The decision carries a reservation, that it may be wrong [3, the two rows][1, what is open, item 20].

6.5 The vacuum content

Assumption 5 (the vacuum content is 0, a declaration fixed by nature's clicks). The massless field rests at a declared level everywhere, the vacuum content, the key **rest** of the universe file, derived by no line of the law; declared 0 in the file's integers at $\Gamma = 6,000$.

Every reader's paces at a rest c_{vac} are $\Gamma - 2c_{\text{vac}}$, the same physics at an effective Γ , so the model cannot tell one rest from another and only nature's clicks can. Light reads the vacuum content into its paces and travels at $(1 - 2x)/\sqrt{3}$, while the massless field's own events travel at $1/\sqrt{3}$, so over L Links the two arrive apart by $2xL\sqrt{3}$ intervals at long wavelength (COMPUTED, CLICKS; S.12). The same line holds at every content on the path: light reads the bodies' wells and the massless field's own events read none, so between a source and the Earth gravity's events lead light by light's whole delay through the potentials on the path, months at least for the Milky Way's potential alone, where nature's GW170817, a gravitational wave and a gamma-ray burst 1.7 seconds apart after 1.3×10^8 years of travel, bounds the difference of the two delays below 1.2×10^{-6} of light's and the difference of the two speeds between -3×10^{-15} and $+7 \times 10^{-16}$ [44]: a finding against nature of the first rank (Section 10.3), derived twice from the row that reads nothing, so the measurement bounds the vacuum content only together with it. The vacuum content stays a declaration, 0 in the file, which the model cannot tell from another rest (ASSUMPTION). The two worlds of Section 9 read no rest, since a fringe's spacing and a calibrated phase read none [1, what is open, item 22][3, the vacuum content is 0, a declaration fixed by nature's clicks].

6.6 The world's numbers, and how the universe was chosen

The universe is chosen by one principle stated before any row: one rule closes everything; the families are its inputs, each a band of the rule named by its pair; the units are a choice of scale; and the universe holds as few free numbers as the rule allows (ASSUMPTION) [3, the standing goal; not one number in the engine, and as few in the files as the law allows]. The road to them:

(i) a family is a band of the rule, and its row holds only its two keys; (ii) three exact rotations, the pairs $[1, 2]$, $[0, 1]$ and $[-1, 2]$ (THEOREM, Section 6.1); (iii) the three of Rule3 is the three axes, so the massless holder of the content carries $1 + 3$ parts and every other family is a scalar; (iv) one denominator, every pair written over Γ , a multiple of 6; (v) the pairs from 1, 2 and 3 (ASSUMPTION): $[1, 1]$ for gravity and the charge's field, $[1, 2]$ for the bound charge, and $[2, 3]$, the first pair past the exact bands, for matter, where nature's tests of gravity require den/num within 2×10^{-5} of 1 (Section 8), so the matter pair is a number for experiment to fix (EXPERIMENT); (vi) every divisor 1, under which the universe carried no free number but the mass; (vii) two numbers where nature's scale forced them, the binding field's gapped pair and gravity's divisor E_g , since under every divisor 1 gravity is as strong as the binding (ASSUMPTION, with the reservation of Section 6.4); (viii) the vacuum content by clicks, 0 by GW170817 (Section 6.5).

What remains: two units, Γ and T ; one bound, the integers' width; and four numbers of physics, matter's numerator m , the binding field's pair, gravity's divisor E_g and the vacuum content, with every other divisor 1. The two worlds read a light-only copy of the file with the three holders of the content at the divisor 10^6 , so that light's own well stays under one level. That this universe is the universe is a decision and not a theorem (ASSUMPTION); the numbers that would close it toward nature, the fine-structure constant, the electron's mass over Planck's and the proton's mass over the electron's, are open and not derived (Section 10.3) [1, the rule's own universe; the final unification][3, the rule's own universe is the universe; the final unification].

7 Bodies

7.1 What a body is, and how it is computed

A body is its record over its Nodes and the quanta that record carries, and never one Node. Its count is the share of its record (Section 4.4), a reading and not a state; its form $D_i \text{ div } T$ is written into the fields it sources, and the fields enter Rule3 through the paces alone. The record is bound by the wells its form sources, by the binding field's screened sum with the range R , while gravity's $1/r$ is weak by E_g and enters no body's binding. No law is written for a body: a body is the support of a standing mode of the one rule, it has no law of motion of its own, and its record moves by Rule3 alone. Four statements make a body (DERIVED from the line, with the decisions named): (a) the body is the fixed point of its binding field, standing where the sources return themselves; (b) a field's level is the rest the start solves from the bodies' forms, after which Rule3 steps it; (c) a body's quanta are its record's share, carried wherever its record's current carries it; (d) a free packet or a cloud falls as $a = \Delta |d\omega_i/dc| \nabla c$, with Δ the band's curvature at $k = 0$, and a compact body by its tail alone, more slowly [1, what a body is; the four lines of the body][3, what a body is].

How a body is computed. By the rule's own iteration and no other formula: start from a spread of the declared count over the Nodes; write the wells that spread lays into the fields and let them rest; find the record's top mode in those paces by Rule3's own read act, iterated; scale its two levels so that the share carries the declared count; write the wells that record lays; and repeat until the levels repeat, in integers, at the fixed point of the division act. The branch is chosen by the start's width alone: a wide start reaches the extended body, the cloud, and a narrow one the compact pixel, two fixed points at one count. A check at the start compares the declared count with the share the laid record reads [1, the generator; the four lines of the body][3, the generator is Rule3].

7.2 Stability

Let a body of M quanta have the record ϕ over its region, $\sum \phi_i^2 = 1$, with the counts $c_i = M\phi_i^2$. Let every field r it sources be a hollow ($\sigma_r = +1$) or a hill ($\sigma_r = -1$), with the static kernel $\mathbf{G}_r$ of its line and the divisor E_r , so that the well at a Node is $\Phi_i = \sum_r \sigma_r (M/E_r) \sum_j G_{r,ij} \phi_j^2$. To first order in Φ/Γ the well raises the one-Node rotation $2 \cos \omega$ by $4g\Phi/\Gamma$, $g = (\text{den} - \text{num})/\text{den}$, and the body's fixed point is a critical point on the sphere of

$$F[\phi] = \frac{\text{num}}{3 \text{den}} \sum_{\text{Links}} (\phi_i - \phi_j)^2 - \frac{2g}{\Gamma} \sum_r \sigma_r \frac{M}{E_r} \sum_{ij} \phi_i^2 G_{r,ij} \phi_j^2, \quad (12)$$

the band's pressure less the wells' energy, each well counted once because the record is its own source (Hartree's form). The record is the top mode of Rule3's symmetric form at the paces, and $dF/d\phi = 2\lambda\phi$ on the sphere is the fixed point's

Table 4 The cloud, the wide branch of the binding row's fixed point, along its branch (DERIVED from the fixed point, a reading of the algebra and no click; GAMEBOARD): the total count, the rotation $2 \cos \omega_b$ against the band's top 1.3333, the rms radius in Links, the count at the centre Node, the well and the inertia over the count and over each other, and the fall against a free packet's. An open 41-cube at $\Gamma = 6,000$, matter [4000, 6000], $T = 2^{15}$, under the massless field alone; the counts in the vacuum's share at the Node, which for a cloud is the engine's count to a percent.

Total count	$2 \cos \omega_b$ (top 1.3333)	rms radius	Centre Node	Well / count	Inertia / count	Inertia / well	Fall / free
4,000	1.3509	8.7	2.6	0.985	1.061	1.077	0.92
3,000	1.3437	10.3	1.0	0.991	1.037	1.047	0.95
2,000	1.3383	11.5	0.4	0.995	1.022	1.027	0.97
1,500	1.3359	12.0	0.3	0.996	1.016	1.020	0.98

own equation with $2 \cos \omega_b = 2 \cos \omega_0 - \lambda$ (THEOREM, GAMEBOARD; S.13). Let $\phi_R(x) = R^{-d/2} \phi(x/R)$ be the body dilated, with d the GameBoard's dimension; the pressure is K/R^2 and each field's well energy is W_r/R^{s_r} , with s_r the field's scaling power at the body's size: $s_r = d$ beyond the field's reach and $s_r = d - 2$ within it or for the massless field [1, the functional and the dilation].

Theorem 4 (the stable body, under the dilation and to first order in the well). *At a fixed point of the binding row of width R the virial $2K = \sum_r \sigma_r s_r W_r$ holds, and the second variation of F along the dilation is $R^2 F''(R) = \sum_r \sigma_r s_r (2 - s_r) W_r$. A body is stable only where this sum is positive; it is a minimum of F where further the Hessian on the modes orthogonal to the dilation and the translations is positive.*

Proof. $F(R) = K/R^2 - \sum_r \sigma_r W_r R^{-s_r}$; $F'(1) = 0$ is $-2K + \sum_r \sigma_r s_r W_r = 0$; $F''(1) = 6K - \sum_r \sigma_r s_r (s_r + 1) W_r = \sum_r \sigma_r (3s_r - s_r^2 - s_r) W_r$: Derrick's scaling [45], Pohozaev's identity [46] on the GameBoard [1, the functional and the dilation]. $\square$

The theorem is a statement about the functional of Eq. (12), to the first order in the well, along the one direction of the dilation, with the dilation taken as continuous; it is not a theorem of the integer step on the full lattice, and the Hessian beyond the dilation is open (Section 10.3).

The factor $s(2 - s)$ is $+1$ at $s = 1$, 0 at $s = 2$ and -3 at $s = 3$: a hollow holds the body where its well falls more slowly than the pressure and breaks it where it falls faster, and a hill does the opposite. Three consequences follow. (a) One hollow beyond its reach binds every mass on a chain, is flat along the dilation on a plane, and in a box has one critical point, a saddle: the body disperses beyond it and collapses within it. (b) A massless field holds the body: Poisson's well in a box has $s = 1$, so $F = a/R^2 - c/R$ has one critical point, $R = 2a/c$, a minimum for every mass, which is Pekar's body [47]. Beside it a hollow beyond its reach adds a barrier, $F = a/R^2 - b/R^3 - c/R$ on a Gaussian body, with the wide critical point a minimum and the narrow one a saddle, and none where $3bc > a^2$, the collapse (COMPUTED, GAMEBOARD). (c) The horizon ends a collapse: beyond first order the well saturates and the Link's pace weakens the pressure, both vanishing together at the pace's floor, so a collapse ends where a Link's pace reaches 0, in a frozen cluster of horizon Nodes [1, the functional and the dilation].

7.3 The two branches: the cloud and the compact pixel

The fixed point of the binding field has two branches at every count above an edge, and the body of nature is the wide one (DERIVED from the fixed point and computed, GAMEBOARD; the identification is a decision of the model) [1, the two branches][3, the body of nature is the cloud].

The cloud, the body of nature. The wide branch is the continuum's Pekar body [47]: the count is spread over many Nodes, with a radius of about $2\Gamma/C$ Links for a count C in the continuum's scaling; the computed clouds of Table 4 are box-limited and their radii grow more slowly. Its three masses agree to the order $(C/\Gamma)^2$: the well it sources, the sum over its Nodes of $D_i \text{ div } T$; the count it resists with, its inertia from the momentum reading; and the weight it falls with, from the momentum identity's force term in a content gradient. It falls as a free quantum falls. Table 4 gives the three along the branch on an open 41-cube at $\Gamma = 6,000$, with the counts in the vacuum's share at the Node, which for a cloud is the implementation's count to a percent [1, the rows against nature, (u)].

The two lighter clouds lean on the box's faces, so their true radii are larger and their numbers nearer 1 than Table 4 shows. So the universality of free fall [36] holds in the model for extended bodies to the order $(C/\Gamma)^2$, and the better the larger Γ is against the body's count. In integers a cloud is sparse: at 1,500 quanta in all there are 0.3 quanta at the densest Node, and the record shared by ten thousand Nodes carries them.

The electron and the proton. The electron is one quantum of the matter family laid with a sense, a message and not a body, stable because one quantum does not disperse into fractions. The proton is a cloud of 1,836 quanta, so $m_p/m_e = 1,836$ is a ratio of counts and fixes no Γ ; the model does not derive it, and the proton's count is an input,

declared in the world file as every body's count is (ASSUMPTION). A body's mass is a click mass: its inertia is read by its click period and de Broglie's k , its weight by the fall's clicks, its well by another body's orbit and its rest energy by the sum line's photons, while the count is the GameBoard's number and enters nature only through the clicks that read it; nature's proton mass is such a click mass, its quarks' rest no part of the comparison [1, the three masses of a body][3, only clicks against clicks] [1, the electron needs no band of its own; the rows against nature, (t)].

The lightest body, and the binding range. Under the binding field a cloud stands only within the range R : the field holds no cloud lighter than about 1.5 times $2\Gamma/R$, so there is a lightest body, the lightest baryon, and the electron is never a body. The cloud of 1,836 is not bound under $R = 10$ or 15 and is bound under [2400, 2401], $R = 20$, with $2 \cos \omega_b = 1.33394$, an rms radius of 11.9 Links, its well 0.997 and its inertia 1.011 of its count and its fall 0.988 of a free packet's. So the file's pair is [2400, 2401], and the range is fixed by the proton (COMPUTED, GAMEBOARD) [1, the rows against nature, (w)].

7.4 A body's clicks

What a detector reads of a body follows from the body's record at its Nodes, the readings of Section 4.3 and the paces, with no further line; every formula below is a derivation with its status and its fence, and none is a run's result [3, the mathematics of the bound bodies and how it bears on the clicks].

A steady rotation writes no light. A body at rest with the plane record $\text{re} = A \cos \omega_b t$ and $\text{im} = eA \sin \omega_b t$, e its ellipticity, has the Wronskian $W = -eA^2 \sin \omega_b$ at every interval, so the write of the holder of the sign is a constant source whose field stands at its rest: no travelling events, no click anywhere. A body in uniform motion radiates nothing either, since its retarded level follows it. A breathing body, $A(t) = A_0(1 + \delta \cos \Omega t)$, has the Wronskian $-eA_t A_{t-1} \sin \omega_b$, which to the first order in δ and for a slow breathing, $\Omega \ll 1$, is $-eA_0^2 \sin \omega_b (1 + 2\delta \cos \Omega t)$, so the light born carries the breathing's frequency Ω and not the rotation ω_b , with $2\delta eA_0^2 \sin \omega_b / T$ quanta per interval in its first harmonic, and a far detector's click rate is that current's share through its front. So a symmetric body at rest clicks nowhere, and a body is seen by what it absorbs and by what it writes when it breathes or moves (DERIVED; the Wronskian and the source GAMEBOARD, the far detector's rate CLICKS) [1, light is born by the write].

The invariant, and the count's drift. For one Node of a body, $a_{\text{next}} + a_{\text{before}} = 2 \cos \omega(t) a_{\text{now}}$ with ω changing slowly has the solution $a_t = C \cos(\sum \omega + \phi) / \sqrt{\sin \omega(t)}$, so $A^2 \sin \omega_b$ is invariant, the discrete adiabatic invariant of Rule3's line, while the form $D = A^2 \sin^2 \omega_b$ and the share are not; the recurrence stepped over 20,000 intervals with ω lowered from 0.841 to 0.600 keeps $D / \sin \omega$ to 4×10^{-5} while D falls by a quarter. The share in quanta at a Node, $(3\Gamma^2 / \sum_a p_a^2) A^2 \sin^2 \omega_b / T$ (Section 4.4), therefore moves under a moving well as

$$\frac{N(t_2)}{N(t_1)} = \frac{\sin \omega_b(t_2)}{\sin \omega_b(t_1)} \cdot \frac{\sum_a p_a^2(t_1)}{\sum_a p_a^2(t_2)}, \quad (13)$$

with no current through any Link: the first factor is the rotation's own change under the clock's pace, the second the Link paces' weights in the share (DERIVED for the invariant, to the first order in the slow variation, and for the drift; GAMEBOARD; S.14). The invariant is what nature calls the number of quanta, E/ω , and the share is nature's energy; the same identity is Planck's relation on the lattice, $\text{share} = I \sin \omega$ with $I = A^2 \sin \omega / T$, so that $E = \hbar \omega$ holds with T for $\hbar$ and $\sin \omega$ in place of ω . The credit counts the share, so a detector of the model credits two beams of equal power the same number of quanta, where nature's photon counter credits the longer wavelength more, by ω_1/ω_2 : the model's unit is a declaration, and the difference against nature's counter is a finding (Section 10.3) (DERIVED; the unit ASSUMPTION; CLICKS) [1, the count is the record's share; the paces' own change moves the share with no current].

The clock's click rate. A bound body's rotation reads the content once: at rest $\cos \omega_0(U) = \cos \omega_0 + 2gU$ to first order, with $g = (\text{den} - \text{num})/\text{den}$, so its rate falls by $f(\omega_0) U$ with $f(\omega_0) = 2(1 - \cos \omega_0)/(\omega_0 \sin \omega_0)$, 1.063 at [2, 3] and 1 as the gap closes; the rate $1 - U$ of a clock that follows the clock's pace alone, p_0/Γ , is the small-gap limit, and a clock that reads the content once is a bound body's rotation and no light clock. A light clock between two faces falls by $2U$, the Link's pace read twice (Section 10.2; S.15); a moving body's tick is the second-order form of Lorentz's factor at its own light speed (Section 8, (e)) (DERIVED, CLICKS).

The resonance. A bound record at the rotation ω_b reads a light level $b_L \cos \omega_L t$ into its content at the weight 1, the content x_0 at its Node otherwise static and the body not yet writing back. At $k = 0$ the Node's line is $a_{\text{next}} + a_{\text{before}} = 2 \cos \omega(t) a_{\text{now}}$ with $2 \cos \omega = 2 - 2g(1 - 2x + 2x^2)$, so the light modulates the rotation by $\epsilon = 4g(1 - 2x_0) b_L / \Gamma$. Writing $a_t = \text{Re}[z_t e^{i\omega_b t}]$ with z slowly varying, the resonant part of the modulation at $\omega_L = 2\omega_b$ gives $z' = -i\epsilon e^{i\vartheta} \tilde{z} / (4 \sin \omega_b)$, whose eigenvalues are real: the amplitude grows at $\mu = \epsilon / (4 \sin \omega_b)$ per interval, the share at 2μ ; detuned by $\Delta = \omega_L - 2\omega_b$ the exponent is $\sqrt{\mu^2 - \Delta^2/4}$, so the body takes light only inside the full width

$\epsilon/\sin\omega_b$. This is the one-mode case; the same modulation mixes the record's frequencies, so a body with two bound modes $\omega_i < \omega_j$ is driven at the sum $\omega_i + \omega_j$, of which $2\omega_b$ is the degenerate case, and transfers between the two at the difference $\omega_j - \omega_i$, the line between two bound states, at the rate $\Omega_R = \epsilon_{ij}/(4\sqrt{\sin\omega_i \sin\omega_j})$ for a modulation ϵ_{ij} of their coupling, the transfer full at resonance and half beyond the detuning $2\Omega_R$, Rabi's line (DERIVED, two derivations) [1, the two-mode line]. The recurrence stepped at $2\cos\omega_b = 1.4317$ confirms the growth (S.16). The read's side is derived; the fraction absorbed, with the body's write back, is open (DERIVED for the resonance, its growth and its width, COMPUTED for the numbers; S.16; the record under the light GAMEBOARD, the body's share rising at its front CLICKS). Beside it: the parametric resonance of an oscillator modulated at twice its frequency, and Rabi's power-broadened line [1, the rows against nature, (y)].

A body's count in clicks. A body's count is its share summed over its Nodes, what a full absorber credits, and under a moving well it drifts by Eq. (13) (DERIVED, CLICKS). The formulas of clicks stand together in Table 2 with their statuses.

7.5 Clusters: how they enter, and what each thing represents

How a cluster enters. The model writes no line for a cluster. One assumption names the universe it describes: every body is bound, no body is free, and a record that is in no body is a message between two clicks, the one that gave it and the one that will take it (ASSUMPTION). A cluster is then bodies standing in one another's wells, bound by the same three fields and clicking at one another; between two bodies there is nothing but the click, the light of the holder of the sign and the two real fields, since no record leaves its body, its tail falling off by $e^{-\kappa}$ per Link, and a quantum crosses a Link whole at a click and nothing else does. A Node is a body's Node or vacuum; it holds its records' levels and remainders and nothing else, and its count is a reading of them (DERIVED, GAMEBOARD) [1, the bound body is one Node; the universe is bound; the algebra of clusters].

The clicks between bodies are clicks in the one sense. Nothing of a cluster changes the definition of Section 5. A body is a region of Nodes with a front, the Ports where its tail meets its neighbour's, so a body is an instrument in the meeting's form: the click between two bodies is the pair (R, t) , one whole quantum credited to the region of the taker at the interval t , drawn by the share Q_R/W_c of the inflow through its front on the forward orbit. An exchange, a join, a parting and a message taken by another body are runs of such clicks, each local and each a credit by the shares; a body's count is its share summed over its Nodes, and it moves only by the clicks at its front, up to the drift of Eq. (13) (DERIVED from the definitions; CLICKS) [3, the click's algebraic definition; the click is the meeting of the future with the past].

What each thing represents in a cluster. (i) *The three levels at a Node are the nesting.* The binding field's level at a Node is the matter's tail, the level of the holder of the sign is the charge's well and the massless field's level is gravity's well: the three kinds of binding by rank, the size of a level the depth in its kind; a reading of the three names the clusters the Node belongs to, a diagnostic and no click (DERIVED, GAMEBOARD). (ii) *Three kinds of binding, not three levels.* Quanta bind into a body by the binding field's screened well, bodies into a charged cluster, an atom or a molecule, by the field of the holder of the sign, and charged clusters into a gravitational cluster by gravity's $1/r$ well; the third kind nests without bound. (iii) *The hierarchy is recursive.* A cluster of clusters is the same structure one level up, its members bodies clicking at one another as a body's Nodes do; no number bounds the depth but the horizon at a Node (Section 3.4). (iv) *The click is the pointer.* The click is the only thing that passes between a cluster and its neighbour and between a level and the next; inside a cluster everything is seen, and beyond its tail only a click is seen. So the observer is a cluster, and what it knows of the world beyond its tail is its clicks, the model's form of the statement that an observer sees only clicks (DERIVED, CLICKS). (v) *The screen is a cluster.* In the rule's own universe everything that stands on the lattice is a body or a cluster of bodies, the source, the screen, a wall, a mirror and a detector alike; the two worlds of Section 9 declare their walls, faces and detectors as instruments instead, each a body's role written without its record (ASSUMPTION). (vi) *The click joins and parts.* Two clusters whose tails overlap read each other by the currents; the cross current of two standing records at the rotations ω_1 and ω_2 is a sum of cosines of $(\omega_1 \mp \omega_2)t$ with no constant term, so at fixed paces no quantum passes at the first order. The click's reach is where the share the overlap sends through a body's front in an interval falls under one quantum, at $d = \ln(a_1 a_2 / T) / \kappa$ Links; beyond it no click is credited and the two are parted. A join and a parting are runs of local clicks, and there is no non-local act (DERIVED; the current GAMEBOARD, the reach CLICKS; S.17). (vii) *The two readings of Γ are two levels.* Every level of the hierarchy has its own clock unit and its own interval, a level's interval being the period of the record one level below it, so the proton to electron ratio is a count's ratio within a cluster and reads no Γ (DERIVED, GAMEBOARD). (viii) *The couplings are at the edge.* Every divisor is 1 but gravity's E_g , the universe of clusters carries Γ and T and nothing else, the body of nature is the cloud, and a stone is clusters of clouds (ASSUMPTION for the divisors, DERIVED for the rest; GAMEBOARD) [1, the levels at a Node are the nesting; three kinds of binding, not three

levels; the hierarchy is recursive; the click is the pointer; the observer is a cluster; the screen is a cluster; the click joins and parts; the two readings of Gamma are two levels; the couplings are at the edge][3, what a body is; the rule's own universe is the universe].

8 The forms of nature the rule arrives at

Each form below follows from the rule's coefficients, read by the geometric optics of a packet's band, before any run; the form of nature beside it is on the comparison side and is never an input. The coefficients contain no metric, no Lorentz transformation, no potential and no force. They contain the paces of Section 3.4, and two features of the paces were chosen with a known result: the factor two on the Link, fixed by light's bending, and the clock's second order, adopted with the perihelion known. Where a form rests on either, it is marked FITTED. $U = c/\Gamma$ denotes the model's potential at the point named, the clock's share per level, and U_b its value at the impact parameter b [1, the rows against nature].

Two potentials. The forms are stated in two potentials. The model's own potential is the clock's share, $U = c/\Gamma$, the content x that the paces read; no instrument reads it. Every test of gravity takes GM from Kepler's law of the orbits of matter and c from light, so it reads $U_K = GM_K/(c^2 r)$. In the model a free packet of matter of the pair [num, den] falls in a content gradient at $a = [(1 - \epsilon)/(3(1 - \epsilon/2))] |\nabla x|$ with $\epsilon = (\text{den} - \text{num})/\text{den}$, exactly from its band, Eq. (4), which with $\cos \omega_0 = 1 - \epsilon$ and $c^2 = 1/3$ is $a = c^2 [2 \cos \omega_0 / (1 + \cos \omega_0)] |\nabla x|$, so the orbits of matter measure

$$U_K = \frac{2 \cos \omega_0}{1 + \cos \omega_0} U, \quad 0.800 \text{ at } [2, 3], \quad 1 \text{ as the gap closes.} \quad (14)$$

Each form below, lettered as the law's rows are, is written first in U , where the two fitted inputs give $\gamma = \beta = 1$ in the post-Newtonian words and the redshift's factor $f(\omega_0) = 1 + \alpha_U$, $\alpha_U = 0.063$ at [2, 3], and then in U_K , where the same algebra gives $\gamma_K = \text{den}/\text{num}$, $\beta_K = (\text{den} + \text{num})/(2\text{num})$ and $1 + \alpha_K = \tan \omega_0/\omega_0$: 1.5, 1.25 and 1.329 at [2, 3], and 1 as the gap closes. The two columns are one algebra, and nature's tests read the second (DERIVED, CLICKS; S.18, S.19) [1, the rows against nature][3, general relativity in the law].

- (a) *The redshift* (DERIVED, CLICKS). Two clocks of one family at the levels c_1 and c_2 tick in the ratio of the root of $(1 - 2U_1 + 2U_1^2)/(1 - 2U_2 + 2U_2^2)$, $1 - (U_1 - U_2)$ to first order, for a band that follows the clock's pace alone; the band's own slope adds a factor, so the rest rotation of a bound record, the clock this row names, shifts by $f(\omega_0)U$ with $f(\omega_0) = 2(1 - \cos \omega_0)/(\omega_0 \sin \omega_0)$, 1.063 at [2, 3] and 1 as the gap closes. Against the Newtonian potential the orbits measure it is $z = (1 + \alpha_K) U_K$ with $1 + \alpha_K = \tan \omega_0/\omega_0$, 1.329 at [2, 3], where nature's clocks give $|\alpha| < 10^{-4}$ [36]. Beside it: Einstein's gravitational redshift, Pound and Rebka's line [48][36].
- (b) *The bending of light* (COMPUTED, FITTED; CLICKS). At a Node of content x light's index is $n = 1/(1 - 2x)$, so $n - 1 = 2x$, because the Link's pace reads the content twice. Far from a body its field is the massless field's rest, $L(r) = 3\sigma_{\text{tot}}/(4\pi r)$, with σ_{tot} the body's wells per interval, and a beam passing at the closest distance b turns by the transverse gradient of the index integrated along the path:

$$\alpha = \int_{-\infty}^{\infty} \frac{\partial n}{\partial b} dl = \frac{3\sigma_{\text{tot}}}{2\pi\Gamma} \int_{-\infty}^{\infty} \frac{b dl}{(b^2 + l^2)^{3/2}} = \frac{3\sigma_{\text{tot}}}{\pi\Gamma b} = 4U_b, \quad U_b = \frac{3\sigma_{\text{tot}}}{4\pi\Gamma b}, \quad (15)$$

twice the angle $2U_b$ that a Newtonian pull of the same potential gives. Computed by the geometric optics of the band, the deflection is 2.02 times Newton's $2U_1/b$, with U_1 the share at one Link from the body, $U = U_1/r$; with the Link's pace lowered once instead of twice it would be 1.01 times Newton's. The factor two is the fitted choice of Section 3.4, and this item checks consistency rather than predicting. Against the Newtonian potential the orbits measure the bending is $(1 + \gamma_K) 2GM_K/(c^2 b)$ with $\gamma_K = \text{den}/\text{num}$, 1.5 at [2, 3], where Cassini's tracking gives $\gamma - 1 = (2.1 \pm 2.3) \times 10^{-5}$ [36][49]. Beside it: Einstein's $4GM/(c^2 b)$ [36].

- (c) *The delay of light* (DERIVED, FITTED; CLICKS; S.20). Light's coordinate speed at the content U is $(1 - 2U)/\sqrt{3}$, so its index is $1 + 2U$, and a signal grazing a body at the impact parameter b between the distances x_1 and x_2 , with $U = \mu/r$, is late by $2\sqrt{3}\mu \ln(4x_1 x_2/b^2)$ intervals: Shapiro's delay with $\gamma = 1$, the same integral whose transverse derivative is the bending, from the same factor two. Against the Newtonian potential the orbits measure it carries the same γ_K . Beside it: Shapiro's delay [50].
- (d) *The perihelion* (COMPUTED, FITTED; CLICKS). By the same geometric optics the matter pair's periapsis advances by $7\pi U_1/(a(1 - e^2))$ per orbit with the clock to first order and by $6\pi U_1/(a(1 - e^2))$, Einstein's value, with the clock's second order, at the semi-axis a and the eccentricity e ; the second order was adopted with this result known. Against the Newtonian potential the orbits measure the advance is $(2 - \beta_K + 2\gamma_K)/3$ of Einstein's, 1.25 at [2, 3] (1.26 on a computed orbit), where the planets' and the Moon's orbits give $|\beta - 1| < 10^{-4}$ [36]. Beside it: Einstein's $6\pi GM/(c^2 a(1 - e^2))$ [36].

- (e) *The moving clock: Lorentz's factor to the second order* (DERIVED, CLICKS; S.21). A packet moving at the velocity v has its tick lengthened by the dispersion of its own band. The tick's factor is $f = \Omega(k)/\omega_0$, with $\Omega(k) = \omega(k) - k\omega'(k)$ the rotation at the moving centre, the phase rate less k times the group velocity, and $\omega(k)$ the band of Eq. (4). From Rule3's exact band of a free record, $\cos \omega = (\text{num}/(3 \text{ den}))(2 + \cos k)$, the expansion in v gives $f = 1 - v^2/(2c_m^2) - (2\omega_0/\sin 2\omega_0)v^4/(8c_m^4) + O(v^6)$ with $c_m^2 = \omega_0/(3 \tan \omega_0)$, the family's own light speed, 13 percent below light's at the computed pair, a reading of the lattice-scale world, while at nature's matter pair $c_m^2/c^2 = \omega_0/\tan \omega_0 = 1 - \omega_0^2/3$ is within 10^{-26} of 1, where the bounds on a difference of matter's and light's limiting speeds are 10^{-15} to 10^{-20} [51][52], no number of the computed pair being nature's: the second-order form of Lorentz's factor at c_m , exactly, and at the fourth order Lorentz's term times $2\omega_0/\sin 2\omega_0$, 1.69 at [2, 3] and 1 as the gap closes. So the full factor is reached only in the limit of a small rest rotation, the limit in which the post-Newtonian parameters reach 1 too; a bound body moves on its own band, whose rest rotation and curvature the generator computes, and carries the same form at its own c_b . It is read as the mean of the front and back click periods $P_v(1 \mp v/u)$ at two resting detectors, with u light's group velocity. No boost is among the 48 symmetries, so Lorentz's factor appears as a property of the clicks and not as a symmetry of the lattice. Nothing here was fitted. Beside it: Lorentz's factor, at the family's own light speed rather than the lattice's [3, the click].
- (h) *De Broglie's relation* (DERIVED; GAMEBOARD for the band, CLICKS for the fringe spacing that reads it; S.22). A free record of a matter family has the band $\cos \omega_b = (\text{num}/(3 \text{ den}))(2 + \cos k)$, the group velocity $v = (\text{num}/(3 \text{ den})) \sin k / \sin \omega_b$, and to second order $k = \omega_b v / c_m^2$ with $c_m^2 = \omega_0/(3 \tan \omega_0)$, the family's own light speed. Beside it: $\lambda = h/p$ with the rest rotation as the mass.
- (i) *The fall* (DERIVED, CLICKS). In a pace gradient a free packet or a cloud accelerates at $a = \Delta |d\omega_0/dc| (c_+ - c_-)/D$ toward the well, with $c_{\pm}$ the read level at its two faces D Links apart. So bodies with the same $\Delta d\omega_0/dc$ fall alike, and a cloud falls as a free quantum falls to the order $(C/\Gamma)^2$ (Section 7.3), while a compact pixel falls by its tail alone, more slowly. The universality holds within one family, and the model has one family of matter. Beside it: Galileo's fall and the equivalence principle [36], which hold in the model for extended bodies and not for collapsed matter.
- (j) *Newton's pull and orbits* (DERIVED, CLICKS). The massless field rests at Poisson's solution with the sources and at Laplace's outside a body, $c(r) = 3s/(4\pi r)$, so the acceleration falls as $1/r^2$ and a circular orbit's v^2 as $1/r$. Beside it: Newton's pull and Kepler's fall-off.
- (m) *A free record keeps its wave number* (THEOREM, GAMEBOARD). Rule3 is invariant under translation, so a free record keeps its wave number and rotation exactly, and a radiating body's period is read unchanged at every distance. Beside it: Hubble's law as Doppler clicks of matter flying apart through the static lattice, with no expansion of space, which the model does not give from the vacuum at any Γ (Section 10.3).

What goes in and what comes out. The model puts in one rule, the paces' read and the files' numbers. In the model's own potential three of the forms above are consistency checks of choices made with them known: the bending (b), the delay (c) and the perihelion (d). Against the potential that nature's tests read, the same three give γ_K , β_K and α_K whose observables, the bending, the perihelion and the redshift, leave Einstein's by 25 to 33 percent and γ_K itself 1 by 50 percent at the rule's own matter pair, reaching 1 only as the matter pair's gap closes; nature's bounds require $\text{den}/\text{num} - 1 < 2 \times 10^{-5}$, so the matter pair is a number for experiment to fix (DERIVED, CLICKS). At the rule's own matter pair the model therefore does not describe the gravity of the solar system; it would in the limit of a closing gap, and whether the model's bodies, the binding range and the proton's cloud stand at a matter pair within 2×10^{-5} of light's is not shown in this paper. Three forms follow without any fitted input: the redshift's band factor (a), the second-order form of Lorentz's factor at the family's own light speed (e) and the universality of free fall for extended bodies (i). The force between moving charges and the sign of the electric force are not derived under the rule as it stands (Section 10.3).

9 The two slits and Bell: the one experiment and the gate, derived and read by the gates

The paper carries no experiment against nature: the results of the two experiments below are derived from the rule and read by the engine's gates (Section 3.7), and neither is performed in a laboratory. No number from a run stands here; what stands is two worlds of the implementation, each as the model's own derivation of what the world must give, marked DERIVED or COMPUTED and never a result. The two slits is the one experiment, the world whose number is Rule3's own propagation on the lattice, its fringe row fixed by the rule's line in real arithmetic and its arrival interval against a continuum wave's the lattice's own number; it is also the implementation's gate, a world whose every number is fixed in advance by the algebra, so that a run can only agree with it to the rounding or reveal a defect of the implementation. Bell's world is a gate and not an experiment: its number is fixed by the declared credit and the parts'

determinism. Each expectation is written before its run and never changed after it, and nothing in a world is built to bring the right result [3, the paper; one experiment and one gate; nothing in an experiment brings the right result by construction].

9.1 The two slits: the world

The emitter is a declared message, a packet of light, the real record of the holder of the sign. Its wavelength is 8 Links along $+x$, $k = \pi/4$, with $\cos \omega = 0.9024$, a period of 14.1 intervals and a group velocity of 0.547 Links per interval. Its levels are laid as $\text{now}_i = b e_i \cos(kx_i)$ and $\text{before}_i = b e_i \cos(kx_i + \omega)$ with $b = 1,328$, about ten quanta per Node, under a raised cosine along x over $x = 4$ to 16, flat across $y = 4$ to 44. The wall is the inner face at $x = 20$ with two gaps of three Nodes, $y = 17$ to 19 and 29 to 31, $d = 12$ Links apart. The GameBoard is flat, $45 \times 48 \times 1$ with the third axis folded, its y faces open and its x faces receding, so that the screen, the last column $x = 44$, $L = 24$ Links from the wall, is opaque and the wall's return through the source's face never re-enters. The screen is twelve declared detectors of four rows each, the smallest region the model allows at this wavelength; the run is 130 intervals, read over the intervals 45 to 120; the universe file is the light-only one. For a record uniform along the folded axis, light's band on the flat GameBoard is the three-dimensional band at $k_z = 0$, so the folded GameBoard carries the world's band exactly (DERIVED, GAMEBOARD). On this world the law's line gives 278.0 quanta through the screen, the peak of the screen's inflow at the interval 68.0 against a continuum's 61.8, the centroid at 72.8 against 68.4, and the wings 16.4 and 12.4 against 23.7 and 20.4 (COMPUTED, CLICKS) [1, the rows against nature, (g)][3, the two slits; one experiment and one gate].

9.2 The two slits: what the model says it must give

Three readings of the one world, each written before any run. (a) *The fringe*. The far field's spacing is $\lambda L/d = 16$ Links; at $L = 2d$ the fringe is the near field's, with the minima at the rows 15 and 33 (DERIVED, CLICKS; S.23). Light's line at the vacuum's paces, $a_{\text{next}} = (\text{the six arrivals})/3 - a_{\text{before}}$, stepped in real arithmetic on the world with its source face open gives 280.8 quanta through the screen, of which 2.8 return through that face, per region from the row 0: 20.5, 19.7, 27.1, 16.2, 9.4, 41.1, 47.7, 14.6, 12.2, 27.1, 21.6 and 23.4 (COMPUTED, CLICKS); the Huygens sum of the two gaps gives 273 with the same central fringe, and the regions 3 and 8 are the lattice's own. The implementation's integers must agree with the real line to the rounding, about a third of a quantum per region; a larger difference is a defect of the implementation and never a finding. (b) *The clicks*. The screen's shares rounded to whole quanta are the expectation, and the row drawn by the shares at the declared seed is the clicks, with the draw's scatter $\sqrt{Np(1-p)}$ per region; the central maximum stands above its two minima by 5.7 standard deviations of that scatter. (c) *The lattice's own number*. The screen's inflow peaks at the interval 68.0 on the lattice against 61.8 for a continuum wave in the same geometry, and the click line resolves the difference (COMPUTED, CLICKS). Under the meeting's credit nothing changes here: light is one real line with nothing to pair, and the share is the record's own inflow [1, the rows against nature, (g)][3, the click is the meeting of the future with the past].

9.3 Bell: the world

In the model's words, entanglement is one event with two trees. The pair is laid as the pair family, two real lines of light never summed at a Node, with equal parts, from one symmetric source, one beam to each side. Each side's detector is a region at least half a wavelength across that reports the two parts' signed level sums at its ports in each interval and knows nothing of angles. The reader applies the declared setting, a pair of integers (p, q) , the instrument's coefficients of Section 5.2, with the port coefficients $e(+) = (p, q)$ and $e(-) = (-q, p)$, exactly orthogonal, so each side's marginal is one half for any setting. The settings of the Clauser-Horne-Shimony-Holt test are $(1, 0)$, $(1, 1)$, $(12, 5)$ and $(5, 12)$, at 0, 45, 22.62 and 67.38 degrees, with no cosine and no root in the implementation. The credit is the one form of the click: the joint share pairs the parts by their label across the two sides through the root,

$$J(p_A, p_B) = \left| \sum_k c_k e_k(p_A) \alpha_k(p_A) e_k(p_B) \beta_k(p_B) \right|^2, \quad c = (1, 1), \quad (16)$$

with α_k and β_k the part k 's amplitude at the port on each side and $e_k(p)$ the port's integer coefficient on the part k , accumulated over the window on both members of the level pair. The joint share is never the square of a local sum, which one detector could form alone and which gives S at most 2. One draw is made per pair: the first side from its marginal and the partner from the conditional (ASSUMPTION, CLICKS) [3, Bell in nature's own layout].

9.4 Bell: what the model says it must give

Three layers, two from the algebra of the GameBoard and the third a declaration. (1) Rule3 gives each part's amplitudes at the ports: the two parts of one beam, laid equal, are two records of one family stepped by the same line along the same path, identical bit for bit at every Node and interval (THEOREM, GAMEBOARD). (2) The meeting with the linearity of Rule3 gives the joint amplitude: the realised click at A's port weights the parts by A's analyser, is read back to the root where the two parts were one event and forward to B, where B's analyser weights them, and the root sums the two branches, $\sum_k c_k e_k(a) e_k(b) = \cos(a-b)/\sqrt{2}$ for equal parts (DERIVED, CLICKS). (3) That the share is the density of the meeting, its square, is Born's rule for a pair [42]: nothing in Rule3 forces it, and it is the least non-local credit that reaches nature's correlations with every quantum counted (ASSUMPTION, FITTED; CLICKS). From the three, with the pair's common phase θ uniform over the pairs and A's + port at a and B's at b ,

$$\begin{aligned}
\text{the meeting:} & \quad P(++) = \frac{1}{2} \cos^2(a-b), \quad E(a,b) = \cos 2(a-b), \quad S = 2\sqrt{2}; \\
\text{a local credit by the shares:} & \quad P(++) = \frac{1}{4} + \frac{1}{8} \cos 2(a-b), \quad E = \frac{1}{2} \cos 2(a-b), \quad S = \sqrt{2}; \\
\text{a local credit by the sign:} & \quad E = 1 - 4|a-b|/\pi \quad (|a-b| \leq \pi/2), \quad S = 2; \\
\text{the credit by the contrast:} & \quad E = \cos 2(a-b) \text{ at the efficiency } 2/\pi, \quad S = 2\sqrt{2},
\end{aligned} \tag{17}$$

The second line follows by averaging $\cos^2(a-\theta) \cos^2(b-\theta)$ over θ , the third from the band of width $|a-b|$ in each half-period where the two signs differ, and the fourth from the integral of $\cos 2(b-\theta)$ over the half-period where $\cos 2(a-\theta) > 0$, against the mean weight $(1/\pi) \int |\cos 2u| du$ (DERIVED, CLICKS). So $E(a,b) = \cos 2(a-b)$ under the meeting. For the declared pairs it is exact, Lagrange's identity of the reader on equal parts: $E = ((p_A p_B + q_A q_B)^2 - (p_A q_B - q_A p_B)^2) / ((p_A^2 + q_A^2)(p_B^2 + q_B^2))$, so at the four settings $E = 119/169, -119/169, 120/169$ and $120/169$ and $S = 478/169 = 2.8284$ exactly, with every pair credited on both sides, an efficiency of 1 by construction and the marginal one half on each side whatever the partner's setting (COMPUTED, CLICKS; S.10). For parts of unequal amplitudes, with r the ratio of the parts' products at the ports, the same credit gives $E(a,b) = \cos 2a \cos 2b + \rho \sin 2a \sin 2b$ with $\rho = 2r/(1+r^2)$, the marginal $(\cos^2 a + r^2 \sin^2 a)/(1+r^2)$ independent of b , and $S = \sqrt{2}(1+\rho)$ at the ideal angles, the two-photon correlation of a pair with the weights $(1,r)$ (DERIVED; ρ and r GAMEBOARD, E and S CLICKS). The maximum of S over the four settings for $E = \cos 2a \cos 2b + \rho \sin 2a \sin 2b$ is $2\sqrt{1+\rho^2}$, Tsirelson's bound $2\sqrt{2}$ at $\rho = 1$ and never the algebraic 4, and the declared pairs' $478/169 = 2.82840$ sits below $2\sqrt{2} = 2.82843$ by the rational angles (DERIVED, CLICKS).

What the GameBoard gives without the declaration is the second and third lines of Eq. (17), two theorems about local credits on the same world (THEOREM, CLICKS): a credit of each side by the shares of its ports gives $S = \sqrt{2}$, and a credit by the sign of its larger port gives the sawtooth, whose sum is exactly 2, Bell's bound. A two-port detector that credits by its contrast and counts nothing where its two ports' shares are nearly equal gives $E = \cos 2(a-b)$ at the efficiency $(2/\pi)V$, the detection loophole of Bell tests [27][28]. Bell's value in this model therefore sits in the credit rule, which is a declaration, and it is no independent evidence that the rule accounts for entanglement. Nature beside the derivation: $S = 2.42 \pm 0.20$ in the loophole-free test of 2015 that measured the sum [30], and the two photon tests of that year [31][32]; 2.42 is what quantum theory predicts for that experiment's state fidelity, $2\sqrt{2}$ the ideal state's. The same credit on four lines gives the three-quantum GHZ correlation $E_3 = \cos 2(a+b+c)$, every pairwise E 0 and Mermin's $M = -4, |M| = 4$ where every local credit gives at most 2, derived by two hands and read by the engine's gate exactly, the blind first (DERIVED under the declared credit, CLICKS) [1, the GHZ gate]. In the world, the two parts of a beam are one record stepped twice, so the lattice enters J only through the common factor $|\alpha\beta|^2$ and E is the Lagrange identity of the ports' coefficients, Malus's law on a declared two-component vector; the gate's run checks the normalisation and the parts' equality and tests no physics beyond the credit. No signal passes between the sides, since the detectors read locally and each side's marginal is one half; no settings are chosen by the GameBoard's own past, so a superdeterministic world [33] is not this one [1, the rows against nature, (h)][3, the click is the meeting of the future with the past; Bell in nature's own layout].

10 Discussion: the constants, the calibration and the findings

10.1 The constants

The causal bound is one Link per interval, and light's own speed at long wavelength is $1/\sqrt{3}$ Links per interval (Section 3.3). The quantum of action is one click, whose energy is the quantum's norm T ; the quantum of distance is the Link and that of time the interval. Newton's constant is not declared: a body whose wells are s quanta per interval makes the massless field's rest $3G(r)s$ at r Links over the field's divisor, $3s/(4\pi r)$ far away, and the level slows the

clock by the potential $\Phi = -c/\Gamma = -U$, Newton's sign, so (DERIVED, GAMEBOARD; S.25)

$$\frac{G_{\text{clock}}}{c^2} = \frac{3}{4\pi\Gamma E_g}, \text{ that is } G_{\text{clock}} = \frac{1}{4\pi\Gamma E_g} \text{ per quantum of well, in Links and intervals,} \quad (18)$$

a reading of Γ , the file's E_g and the lattice. With $c^2 = 1/3$, $G_{\text{clock}} = 1/(4\pi\Gamma E_g)$ is the number per quantum of well that lowers the clock's share, and Kepler's G , read from the orbits of matter as clicks, is $[2 \cos \omega_0 / (1 + \cos \omega_0)] G_{\text{clock}}$ by Eq. (14), 0.8 of it at $[2, 3]$ (DERIVED, CLICKS). The vacuum content, the massless field's declared rest, is the model's cosmological constant, 0 by nature's clicks (Section 6.5) [1, the constants].

10.2 Nature's numbers enter only as clicks

Every measured number of an experiment is first written as a click pattern on a declared detector against a declared clock. The families' pairs and the units Γ and T , what in nature is one Link and one interval, are fixed by that match and by nothing read off the GameBoard; the two ratios the model fixes without it are light's speed of $1/\sqrt{3}$ Links per interval and the causal bound of 1 (ASSUMPTION, CLICKS) [1, the reading; nature's numbers enter the board only as clicks]. Every formula of Section 8 is derived in three steps: the law's line in the GameBoard's words; each word traded for a detector's report of it, ω for Planck's share or a clock's rate, k for a fringe's spacing, $A^2 \sin^2 \omega/T$ for the count $N = Q_R/W_c$, U for the ratio of two clocks' rates, m^* for a fringe's spacing at a known speed, a speed for two arrival times; and what remains, a relation among click numbers alone, set beside nature's [3, only clicks against clicks]. The calibration series, one number of the model against one click pattern of nature per item:

- (1) *The two speeds.* Gravity's own events and light at a content x arrive apart by $2xL\sqrt{3}$ intervals over L Links (Section 6.5); GW170817 bounds gravity's lead over light at 7×10^{-16} of its speed and the difference of the two delays through the path's potentials at 1.2×10^{-6} of light's [44], which the row that reads nothing does not meet (Section 10.3); the vacuum content is declared 0. Gravity's divisor is bounded from below by light's own fringes: where a light packet's own well reached one level its fringes would move with the intensity, which nature's fringes never do, so E_g is at least the packet's summed form over T for the strongest light measured, and the light-only worlds use 10^6 at $\Gamma = 6,000$ (DERIVED, CLICKS).
- (2) *Light's dispersion on the lattice.* Along an axis the group speed falls as $1 - 0.083k^2$. The photons of one gamma-ray burst arrive together up to the TeV range, which bounds a quadratic dependence of light's speed on its energy: MAGIC's limit on the scale of such a dependence, $E_{\text{QG},2}$ above about 6×10^{10} GeV [53], bounds $|\delta v/v| = (3/2)(E/E_{\text{QG},2})^2$ below about 4.5×10^{-16} at $E = 1$ TeV (Fermi's linear bound is far tighter and does not apply to a quadratic form [54]), so the Link is below about 10^{-26} metres, the first bound on the unit of Γ (COMPUTED, CLICKS; S.24).
- (3) *Matter's two slits.* A matter packet's fringe spacing $\lambda L/d$ and a time of flight between two detectors give its inertia m^* in Links and intervals, and nature's $h/(mv)$ with the units of (1) and (2) gives the matter pair.
- (4) *The light clock.* A packet between two mirror faces L Links apart ticks with the period $2L\sqrt{3}/(1 - 2x)$ intervals in a static well of content x , its rate falling by $2x$ where a bound body's rotation falls by $f(\omega_0)x$, the finding of Section 10.3 (DERIVED, CLICKS).

None of the numbers is fixed today, the vacuum content being declared, and two are bounded, gravity's divisor and the Link. Until the rest are named, run and read, the model's numbers are its own and not nature's, and the two worlds of Section 9 show that the implementation works and are not these experiments.

10.3 Findings against nature and open questions

Findings against nature. The quantum of credit is an energy, T per click whatever the frequency, so a counter of the model credits two beams of equal power alike, where nature's photon counter credits the longer wavelength more and the photoelectric effect reads one click per quantum with the energy proportional to the frequency: the model's click is not Einstein's photon (Section 7.4). The proton is a cloud at least the electron's Compton length across, $0.577/\omega_0$ Links, where nature's proton is 460 times smaller than the electron's Compton length, and a heavy family small enough for a compact proton would break the universality of free fall; the binding holder, read by the pace as gravity is, couples two quanta at $\alpha_b = (3\sqrt{3}/4\pi)\omega_0 f(\omega_0)/\Gamma$, below 10^{-27} at nature's numbers, so the model's strong force as the two rows have it binds no nucleus at nature's scale, against the mass defect of 8 MeV per nucleon and the form factor's 0.84 fm: the proton's size and nuclear binding are open by name, a second matter family and a binding read of the rotation's kind the hypotheses beside them (DERIVED for the coupling, CLICKS). The massless field reads no content, so its events run at $1/\sqrt{3}$ through every well while light runs at $(1 - 2U)/\sqrt{3}$, against GW170817 (Section 6.5); a row that reads the content as every family does would close it and is a hypothesis by name, its second order owed first. Matter flying apart through the static lattice under the Newtonian potential the orbits measure obeys Friedmann's dust

equation, Hubble's law and the supernovae's time dilation as Doppler clicks with no expansion of space, and decelerates, $q_0 = \Omega_m/2 > 0$, where the supernovae read $q_0 = -0.55 \pm 0.1$ (DERIVED, two derivations, CLICKS), a vacuum content below zero and a push in the massless row's rest beyond Laplace's the hypotheses beside it by name. The charged cloud is bound a little less than the neutral one by its own hill and is heavier by about 10^{-5} , where nature's neutron, the neutral one, is the heavier by 1.4×10^{-3} [1, the rows against nature, (v) and (z)]. With the diagonal tension alone, the massless field's travelling events carry the breathing polarisation and the lattice-aligned + polarisation and no $\times$, where nature's gravitational waves carry + and $\times$ at every orientation; and light's band is anisotropic, in the phase by $k^2/48$ between an axis and a face diagonal and by $k^2/36$ between an axis and the body diagonal, where light is dispersionless, about 1 percent at a wavelength of 8 Links and falling as the square of the Link over the wavelength, which bounds the Link from nature's isotropy [1, what is open, item 23]. The law has no current's read for real records, and gravity's readers are real records, so a neutral gyroscope precesses by 0 from a rotating body's motion, where Gravity Probe B measured 37.2 ± 7.2 milliarcseconds per year against general relativity's 39.2 [55]; closing this would need gravity's readers to carry a phase, which the law does not have (DERIVED, CLICKS) [3, general relativity in the law; the current's read]. And against the Newtonian potential the orbits measure, the post-Newtonian observables leave Einstein's by 25 to 33 percent and γ_K 1 by 50 percent at the rule's own matter pair, where nature's pair, the closing gap, holds no computed body (Section 8). No cluster of Section 7.5 is compared with nature in this paper.

Open. The magnetic force between moving charges, derived under the rotation as the Link's phase and not run; the finding by name of the like-or-unlike look, that two charged bodies' response to their own angle is not additive against a neutral background; an exactly conserved integer charge, the remainder term of Section 5.5; the unit of charge; the units Γ and T and gravity's divisor E_g , which wait on nature's clicks (EXPERIMENT); the matter pair's gap, which nature's tests bound, and whether the model's bodies stand at a pair within 2×10^{-5} of light's (Section 8); what fixes Γ and $\sqrt{T}/\Gamma$; the fine-structure constant and the mass ratio; the hierarchy of gravity and electricity, held with a reservation (Section 6.4); the stability theorem beyond the dilation; whether a breathing body cools; which of a bound body's two frequencies, its rest rotation, which redshifts as $f(\omega_0)U$, or the beat between two of its bound modes, which a first derivation finds rising in a well, nature's atomic clock reads, an exact answer being owed before the redshift row is compared with a transition line, and with it the light clock between declared faces, which falls by $2U$ where a bound body's rotation falls by $f(\omega_0)U$, against nature's cavity and atomic clocks agreeing to 10^{-5} and better [36], one finding with the beat's, a bound body's size in Links not shrinking by $1 - U$ in a level, whose mending line must be a dilation of the whole bound problem; a current's read for real records; and whether the tension's write, whose composed booking reaches three Links in one interval, meets the local test as stated (Section 4.3) [1, what is open].

11 Conclusion

One line of integer arithmetic at every Node of a cubic lattice, with one remainder kept at the Node, is reversible to the bit, conserves a quadratic form with its current at fixed paces without the rounding, and carries the momentum flux as the wave's own stress. On that lattice there are only events that split, run forward from an emitter as the future and read backward from a click as the past, and a click is their meeting at a declared detector, the choice among the meetings being the one non-local element. Every formula of the paper is reached from that line by a chain of named steps with its assumptions named, closed in the algebra. Families are bands of the rule, each one row of a file; bodies are its fixed points, and a cluster of bodies is the same structure one level up, with the click the only thing between its members. Gravity is a level that slows a Node's clock once and its Links twice; in the model's own potential two of its features were fitted, to light's bending and to the perihelion, and against the potential that Kepler's orbits measure the same forms reach Einstein's only as the matter family's gap closes, a limit not computed here, so at the rule's own matter pair the model does not describe the gravity of the solar system. Bell's value rests on a credit rule declared to reach nature's correlations and is no independent evidence that the rule accounts for entanglement. The second-order form of Lorentz's factor, at the family's own light speed, exact at that order and reaching the full factor only as the family's gap closes, and the universality of free fall for extended bodies follow without fitting. The findings against nature, listed once in Section 10.3, stand and are not closed, each needing a change of the model or of its identifications; the magnetic force waits on a run under the holder's rotation. The two slits and Bell are stated as what each must give, their results derived from the rule and read by the engine's gates, no experiment against nature performed; what remains is to compare a body's clicks with nature's, in the order one computer allows. We hold the whole to be the least computation we know of, in the three counts named in the introduction, that gives the two slits and Bell under the declared credit, with no theorem of minimality; why nature is so, we do not claim to know.

Statements and Declarations

Funding. None.

Competing interests. None.

Ethics approval, consent to participate and consent for publication. Not applicable; no human or animal participants.

Data and code availability. The law's document, the implementation and the figures' scripts are developed in the public Git repository <https://github.com/Closer24/Universe24> and archived at Zenodo under the concept DOI 10.5281/zenodo.22738746 [41], under the MIT licence. The paper reports the gates' runs (Section 3.7), and the worlds of the two slits and of Bell are in the repository as files.

Author contributions. The author laid down every postulate and decision of the model, directed the work and wrote the paper.

The use of a large language model. The author used a large language model (Claude, Anthropic, through Claude Code, 2026) extensively as a tool, under the author's direction, for the software, the computations, the mathematical derivations and their checks, and the drafting of the text. The model is not an author. The author reviewed every statement and takes full responsibility for the content. No figure was made by a generative tool; every figure is drawn by a script from the definitions or from the law's document.

References

- [1] The law of the GameBoard, docs/ALGEBRA.md of the archive [41], cited by the name of its section: the GameBoard and the families (the objects, a family's declaration, the interval, readings and measurements); Rule3 (the line, the direction, the four acts, the paces, the bound, the self-source, the conserved form); the click (the booking, the count is the record's share, the click is the detector's report, no click names a Node, the algebra of the exclusivity); the primitives; the stable body (what a body is, the well, the functional and the dilation, the surplus leaves, the generator, the velocity); what the law says of nature (the reading, with no waves on the event board, what produces a click is not known and nature's numbers enter the board only as clicks; the hypotheses under their own names; the postulates; the constants; the rows against nature); what is open.
- [2] The engine, docs/ENGINE.md of the archive [41]: the words, the NodeState, the interval, the loader and the files, the output, how to run a world, how to add a family, the gates.
- [3] The decisions, docs/HIGHLIGHTS.md of the archive [41]: the model's decisions in force, one line each, cited by the line's name.
- [4] K. Zuse, Rechnender Raum (Vieweg, Braunschweig, 1969); English translation: Calculating Space, MIT Technical Translation AZT-70-164-GEMIT (MIT Project MAC, Cambridge, 1970).
- [5] E. Fredkin and T. Toffoli, Conservative logic, International Journal of Theoretical Physics 21, 219 (1982), doi:10.1007/BF01857727.
- [6] N. Margolus, Physics-like models of computation, Physica D 10, 81 (1984), doi:10.1016/0167-2789(84)90252-5.
- [7] T. Toffoli and N. Margolus, Invertible cellular automata: a review, Physica D 45, 229 (1990), doi:10.1016/0167-2789(90)90185-R.
- [8] S. Wolfram, A New Kind of Science (Wolfram Media, Champaign, 2002).
- [9] G. 't Hooft, Duality between a deterministic cellular automaton and a bosonic quantum field theory in 1+1 dimensions, Foundations of Physics 43, 597 (2013), doi:10.1007/s10701-013-9702-z.
- [10] G. 't Hooft, The Cellular Automaton Interpretation of Quantum Mechanics, Fundamental Theories of Physics 185 (Springer, Cham, 2016), doi:10.1007/978-3-319-41285-6.
- [11] I. Bialynicki-Birula, Weyl, Dirac, and Maxwell equations on a lattice as unitary cellular automata, Physical Review D 49, 6920 (1994), doi:10.1103/PhysRevD.49.6920.
- [12] D. A. Meyer, From quantum cellular automata to quantum lattice gases, Journal of Statistical Physics 85, 551 (1996), doi:10.1007/BF02199356.
- [13] P. Arrighi, An overview of quantum cellular automata, Natural Computing 18, 885 (2019), doi:10.1007/s11047-019-09762-6.
- [14] T. Farrelly, A review of quantum cellular automata, Quantum 4, 368 (2020), doi:10.22331/q-2020-11-30-368.
- [15] G. M. D'Ariano and P. Perinotti, Derivation of the Dirac equation from principles of information processing, Physical Review A 90, 062106 (2014), doi:10.1103/PhysRevA.90.062106.
- [16] R. P. Feynman, Simulating physics with computers, International Journal of Theoretical Physics 21, 467 (1982), doi:10.1007/BF02650179.
- [17] J. A. Wheeler and R. P. Feynman, Interaction with the absorber as the mechanism of radiation, Reviews of Modern Physics 17, 157 (1945), doi:10.1103/RevModPhys.17.157.
- [18] J. G. Cramer, The transactional interpretation of quantum mechanics, Reviews of Modern Physics 58, 647 (1986), doi:10.1103/RevModPhys.58.647.
- [19] Y. Aharonov, P. G. Bergmann and J. L. Lebowitz, Time symmetry in the quantum process of measurement, Physical Review 134, B1410 (1964), doi:10.1103/PhysRev.134.B1410.
- [20] Y. Aharonov and L. Vaidman, The two-state vector formalism: an updated review, Lecture Notes in Physics 734, 399 (Springer, Berlin, 2008), doi:10.1007/978-3-540-73473-4_13.
- [21] H. Price, Time's Arrow and Archimedes' Point: New Directions for the Physics of Time (Oxford University Press, New York, 1996).
- [22] K. B. Wharton and N. Argaman, Colloquium: Bell's theorem and locally mediated reformulations of quantum mechanics, Reviews of Modern Physics 92, 021002 (2020), doi:10.1103/RevModPhys.92.021002.

- [23] N. Harrigan and R. W. Spekkens, Einstein, incompleteness, and the epistemic view of quantum states, *Foundations of Physics* 40, 125 (2010), doi:10.1007/s10701-009-9347-0.
- [24] J. S. Bell, On the Einstein Podolsky Rosen paradox, *Physics Physique Fizika* 1, 195 (1964), doi:10.1103/PhysicsPhysiqueFizika.1.195.
- [25] J. F. Clauser, M. A. Horne, A. Shimony and R. A. Holt, Proposed experiment to test local hidden-variable theories, *Physical Review Letters* 23, 880 (1969), doi:10.1103/PhysRevLett.23.880.
- [26] A. Fine, Hidden variables, joint probability, and the Bell inequalities, *Physical Review Letters* 48, 291 (1982), doi:10.1103/PhysRevLett.48.291.
- [27] P. M. Pearle, Hidden-variable example based upon data rejection, *Physical Review D* 2, 1418 (1970), doi:10.1103/PhysRevD.2.1418.
- [28] P. H. Eberhard, Background level and counter efficiencies required for a loophole-free Einstein-Podolsky-Rosen experiment, *Physical Review A* 47, R747 (1993), doi:10.1103/PhysRevA.47.R747.
- [29] A. Garg and N. D. Mermin, Detector inefficiencies in the Einstein-Podolsky-Rosen experiment, *Physical Review D* 35, 3831 (1987), doi:10.1103/PhysRevD.35.3831.
- [30] B. Hensen et al., Loophole-free Bell inequality violation using electron spins separated by 1.3 kilometres, *Nature* 526, 682 (2015), doi:10.1038/nature15759.
- [31] M. Giustina et al., Significant-loophole-free test of Bell's theorem with entangled photons, *Physical Review Letters* 115, 250401 (2015), doi:10.1103/PhysRevLett.115.250401.
- [32] L. K. Shalm et al., Strong loophole-free test of local realism, *Physical Review Letters* 115, 250402 (2015), doi:10.1103/PhysRevLett.115.250402.
- [33] S. Hossenfelder and T. Palmer, Rethinking superdeterminism, *Frontiers in Physics* 8, 139 (2020), doi:10.3389/fphy.2020.00139.
- [34] T. Regge, General relativity without coordinates, *Il Nuovo Cimento* 19, 558 (1961), doi:10.1007/BF02733251.
- [35] L. Bombelli, J. Lee, D. Meyer and R. D. Sorkin, Space-time as a causal set, *Physical Review Letters* 59, 521 (1987), doi:10.1103/PhysRevLett.59.521.
- [36] C. M. Will, The confrontation between general relativity and experiment, *Living Reviews in Relativity* 17, 4 (2014), doi:10.12942/lrr-2014-4.
- [37] P. Touboul et al., MICROSCOPE mission: final results of the test of the equivalence principle, *Physical Review Letters* 129, 121102 (2022), doi:10.1103/PhysRevLett.129.121102.
- [38] J. Collins, A. Perez, D. Sudarsky, L. Urrutia and H. Vucetich, Lorentz invariance and quantum gravity: an additional fine-tuning problem?, *Physical Review Letters* 93, 191301 (2004), doi:10.1103/PhysRevLett.93.191301.
- [39] D. Mattingly, Modern tests of Lorentz invariance, *Living Reviews in Relativity* 8, 5 (2005), doi:10.12942/lrr-2005-5.
- [40] R. Courant, K. Friedrichs and H. Lewy, Über die partiellen Differenzengleichungen der mathematischen Physik, *Mathematische Annalen* 100, 32 (1928), doi:10.1007/BF01448839.
- [41] A. Gonen, Universe24: a discrete simulator of local physical laws (Reality Theory), Zenodo (2026), doi:10.5281/zenodo.22738746 (the concept DOI, resolving to the latest version).
- [42] M. Born, Zur Quantenmechanik der Stoßvorgänge, *Zeitschrift für Physik* 37, 863 (1926), doi:10.1007/BF01397477.
- [43] H. Yukawa, On the interaction of elementary particles. I, *Proceedings of the Physico-Mathematical Society of Japan* 17, 48 (1935), doi:10.11429/ppmsj1919.17.0.48.
- [44] B. P. Abbott et al. (LIGO Scientific Collaboration, Virgo Collaboration, Fermi Gamma-ray Burst Monitor and INTEGRAL), Gravitational waves and gamma-rays from a binary neutron star merger: GW170817 and GRB 170817A, *The Astrophysical Journal Letters* 848, L13 (2017), doi:10.3847/2041-8213/aa920c.
- [45] G. H. Derrick, Comments on nonlinear wave equations as models for elementary particles, *Journal of Mathematical Physics* 5, 1252 (1964), doi:10.1063/1.1704233.
- [46] S. I. Pohozaev, Eigenfunctions of the equation $\Delta u + \lambda f(u) = 0$, *Soviet Mathematics Doklady* 6, 1408 (1965).
- [47] S. I. Pekar, *Untersuchungen über die Elektronentheorie der Kristalle* (Akademie-Verlag, Berlin, 1954).
- [48] R. V. Pound and G. A. Rebka, Apparent weight of photons, *Physical Review Letters* 4, 337 (1960), doi:10.1103/PhysRevLett.4.337.
- [49] B. Bertotti, L. Iess and P. Tortora, A test of general relativity using radio links with the Cassini spacecraft, *Nature* 425, 374 (2003), doi:10.1038/nature01997.
- [50] I. I. Shapiro, Fourth test of general relativity, *Physical Review Letters* 13, 789 (1964), doi:10.1103/PhysRevLett.13.789.
- [51] S. Coleman and S. L. Glashow, High-energy tests of Lorentz invariance, *Physical Review D* 59, 116008 (1999), doi:10.1103/PhysRevD.59.116008.
- [52] F. W. Stecker, Limiting superluminal electron and neutrino velocities using the 2010 Crab Nebula flare and the IceCube PeV neutrino events, *Astroparticle Physics* 56, 16 (2014), doi:10.1016/j.astropartphys.2014.02.007.
- [53] V. A. Acciari et al. (MAGIC Collaboration), Bounds on Lorentz invariance violation from MAGIC observation of GRB 190114C, *Physical Review Letters* 125, 021301 (2020), doi:10.1103/PhysRevLett.125.021301.
- [54] A. A. Abdo et al. (Fermi GBM and LAT Collaborations), A limit on the variation of the speed of light arising from quantum gravity effects, *Nature* 462, 331 (2009), doi:10.1038/nature08574.
- [55] C. W. F. Everitt et al., Gravity Probe B: final results of a space experiment to test general relativity, *Physical Review Letters* 106, 221101 (2011), doi:10.1103/PhysRevLett.106.221101.